

Doubts about Consumption Dynamics and Mobility: Welfare Costs across the Distribution*

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Abstract

The welfare cost of model uncertainty depends on its source and amount. A household at a percentile of the consumption distribution may distrust the conditional distribution of consumption innovations, the transition probabilities that move it across percentiles, or both. We assign a separate relative-entropy penalty to each doubt in a nested multiplier recursion and estimate both components from Consumer Expenditure Survey data for 1990Q1–2024Q1, ranking households once by consumption and once by private income. Under expected utility, mobility redistributes welfare costs across percentiles and leaves their long-run average unchanged. Doubts about the transition matrix break that neutrality. At a common detection-error probability, doubts raise the cost of consumption risk to about 0.21–0.22 annual percentage points of consumption growth. At the 10th, 50th, and 90th percentiles, the increment from transition uncertainty is 1.3–1.6 times the increment from innovation uncertainty.

Keywords— Welfare costs, mobility, model uncertainty, robustness, heterogeneity.

JEL codes: E21, E32, D31, D81, C61, C63

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1 Introduction

We extend the representative-household welfare calculations of Lucas (1987, 2003) to households that occupy different percentiles of CEX consumption and income distributions and face risks from consumption dynamics within percentiles and mobility across percentiles. We calculate consumption-equivalent costs at each percentile relative to a benchmark that removes innovations in percentile-mean consumption while preserving mobility. We then study how concerns about model misspecification affect those costs. Households use our statistical model to forecast future consumption but fear that it is misspecified. Following Hansen and Sargent (2001, 2008b) and Barillas et al. (2009), they evaluate consumption under adverse models that are difficult to distinguish from the approximating model in samples of the lengths they have seen. We study how the welfare costs of these doubts differ across percentiles.¹

A household wants to forecast both its future percentile and consumption within each percentile it might reach. Doubts about within-percentile consumption dynamics lead it to place more weight on adverse consumption outcomes within each possible percentile. Doubts about mobility lead it to place more weight on percentiles with low conditional expected continuation values.

In our main calibration, the alternatives lie close enough to the approximating model that the household cannot reliably tell which model generated its observations. We hold the degree of doubt fixed by requiring the alternatives associated with the two concerns to be equally hard to distinguish from the approximating model. The *source* of doubt determines both the size of the welfare cost and its variation across percentiles. Percentile by percentile, we calculate both the cost of consumption risk and the additional cost induced by model misspecification.

We build on the analysis of Sargent et al. (2026a). They modify Lucas’s setup by estimating a linear–Gaussian model of evolving cross-sections of private income, post-tax income, and consumption from the Consumer Expenditure Survey (CEX). They represent the logarithm of consumption at each percentile in the cross-section as an additive functional (see Hansen (2012)) driven by a reduced-rank first-order vector autoregression. They estimate this autoregression using Dynamic Mode Decomposition (DMD) (see Schmid (2010), Kutz et al. (2016), Sargent et al. (2026b)). Their households remain at fixed percentiles and

¹Following a recommendation of Lucas (2003), we ignore information about the welfare costs of risks in consumption growth that is embedded in asset prices according to formulations of Hansen et al. (1999), Tallarini (2000), and Alvarez and Jermann (2004). Barillas et al. (2009) argued that endowing agents with concerns about model misspecification provides a way to overcome Lucas’s objection to using asset prices to calibrate welfare costs of aggregate fluctuations.

evaluate consumption streams with standard expected-utility preferences, without doubting the statistical model. Sargent et al. find that serially correlated risks generate much larger welfare costs than do i.i.d. risks, and that benefits (or costs) from participating in the U.S. tax-transfer-and-insurance system dwarf benefits from eliminating consumption-growth risk. We extend their analysis in two ways.

Our first extension captures mobility across percentiles via a Markov chain with transition matrix Π whose entry Π_{pj} is the probability that a household currently at percentile p moves to percentile j next period. A household’s value function then depends on the percentiles it may occupy in the future. We estimate Π from consecutive CEX interview pairs, ranking households by consumption and by private income. Without doubts, mobility redistributes welfare costs across percentiles and leaves their π -weighted mean unchanged, where π is the stationary distribution of Π . Doubts about Π break that neutrality. The worst-case matrix tilts toward reachable destinations with low continuation values, which raises the mean cost as well as moving it across percentiles.

Our second extension allows households to doubt the statistical models used to evaluate continuation values. We equip each agent with what Hansen and Sargent (2001, 2008b) and Barillas et al. (2009) call multiplier preferences with “multiplier” or “penalty” parameter θ on relative-entropy divergences from an approximating statistical model. Households inside our model share the approximating model with each other and with us, the model builders.² We represent alternative statistical specifications by likelihood ratios and measure departures from the approximating model using relative entropy. A large penalty parameter θ makes relative entropy distortions expensive and so keeps the models a household takes seriously close to the approximating model. As $\theta \rightarrow \infty$, a household’s preferences over consumption plans become equivalent to expected discounted utility preferences.³ To capture the “structured” misspecification concerns studied by Hansen and Sargent (2007, 2022), in which a decision maker distrusts some components of a model more than others, we introduce two penalties on relative entropy distortions, one that controls doubts about consumption dynamics conditional on a next-period percentile and another that controls doubts about transitions across percentiles.

Empirical evidence shows that households’ beliefs about mobility can depart systemat-

²Multiplier preferences are closely related to the stochastic differential utility specification of Duffie and Epstein (1992). In both specifications, the decision maker especially dislikes serially correlated risk. Because our households’ approximating statistical model presents serially correlated risk, we can anticipate that adding concerns about robustness will amplify welfare gains from reducing serially correlated risks relative to the welfare gains computed by Sargent et al. (2026a). In Section 5.1, the persistent component accounts for about 98 percent of the Case 0 cost under consumption ranking and 79–80 percent under private income.

³The Lagrangian correspondence associates each multiplier θ with an entropy-constraint radius (Hansen and Sargent, 2008b, Ch. 7).

ically from estimated transition probabilities. Rozsypal and Schlafmann (2023) show that households overestimate the persistence of their own income process by an amount that depends on where in the distribution they sit, and Alesina et al. (2018) document large and patterned errors in what people believe about the chances of moving through the distribution across generations. Overestimating persistence is pessimism at the bottom of the distribution and optimism at the top, while our worst-case transition matrix is pessimistic everywhere. Under consumption ranking the worst case shifts probability toward low-consumption destinations at every current percentile, and under private-income ranking it erodes persistence at high income ranks (Section 5.3). Section 2.3 gives a separate, statistical reason for such doubts.

Switching the two types of doubts on and off yields four cases:

- *Case 0*: both doubts off. This recovers the discounted-expected-utility value function of Sargent et al. (2026a), extended to allow Markov transitions across percentiles.
- *Case 1*: doubts about consumption dynamics alone (innovation uncertainty).
- *Case 2*: doubts about mobility alone (transition uncertainty).
- *Case 3*: both doubts active.

Following Anderson et al. (2003) and Barillas et al. (2009), we discipline the two penalty parameters with detection-error probabilities. We set each penalty so that an observer who sees one household’s percentile history and the aggregate cross-section over our sample could distinguish the household’s worst-case model from the approximating model only slightly better than by flipping a coin. Section 4.6 explains how the household’s information determines this calibration. Applied to CEX data, joint doubts raise the welfare cost of consumption fluctuations from 0.086–0.105 annual percentage points of consumption growth under expected utility to about 0.21–0.22 (Table 3). In consumption-level units, the median cost rises from 2.12 to 5.39 log percentage points under consumption ranking and from 2.56 to 5.29 under private-income ranking. We find that transition uncertainty alone produces an increment roughly 1.3–1.6 times that of innovation uncertainty at the reported percentiles. When both doubts operate at a common detection-error target, the joint increment is about 59–66 percent of the sum of the two separate ones, because a single detection-error probability must then discipline two distortions at once (Appendix J). Transition uncertainty also carries nearly all of the variation across percentiles. From the 10th to the 90th percentile, its increment declines under consumption ranking and rises under private-income ranking, while the innovation increment is nearly flat under both. We vary the detection-error target from 0.45 down to 0.10, where the cost reaches 1.3–1.5 annual percentage points.

The joint cost of consumption risk remains much smaller than the insurance benefit

reported in Sargent et al. (2026a). There, participation in the U.S. tax-transfer-and-insurance system is worth about four percentage points of annual growth to a household at the 10th percentile.

Related literature

We build first on the literature measuring the welfare costs of consumption fluctuations. Lucas (1987) computed the consumption-equivalent welfare cost of aggregate consumption fluctuations in a representative-agent model, while Obstfeld (1994) and Tallarini (2000) showed how a richer model of consumption growth and recursive preferences can produce substantially larger costs. Like these papers, we express welfare costs in consumption-equivalent units and take a statistical model of consumption itself as the primitive of our analysis.

In spirit, we are closest to work on the welfare costs of fluctuations in economies with heterogeneous households. In incomplete-insurance economies, the welfare cost of aggregate fluctuations also depends on their interaction with persistent idiosyncratic risk. Storesletten et al. (2001) and Krebs (2003) emphasize countercyclical variation in labor-income or human-capital risk. Krebs (2007) emphasizes cyclical variation in the persistent earnings losses that follow job displacement. De Santis (2007) shows that uninsured background consumption risk convexly magnifies the value of eliminating aggregate risk even without countercyclical idiosyncratic volatility. Mukoyama and Şahin (2006) and Krusell et al. (2009) further show that aggregate averages can conceal large differences across skill, wealth, and employment states. These mechanisms can generate consumption-equivalent costs an order of magnitude or more above the estimate in Lucas (1987). Ellison and Sargent (2015) connect this heterogeneous-risk literature to robust preferences. Uninsured individual consumption risk magnifies the increment to fluctuation costs caused by concern about misspecification, and that concern in turn magnifies the contribution of individual risk. We use a household's consumption or income percentile as its discrete state, measure consumption dynamics and mobility directly in the data, and let doubts target these two components separately. A household can distrust the distribution of the innovation ε_{t+1} conditional on its next-period percentile p_{t+1} , the transition probabilities Π_{p_t} that move it across percentiles, or both.

We take our empirical foundation from research using the Consumer Expenditure Survey to study the joint distribution of income and consumption. Cutler and Katz (1992), Attanasio and Davis (1996), Krueger and Perri (2003, 2006), Krueger et al. (2010), and Heathcote et al. (2010) document the evolution of income and consumption inequality and the relationship between the two distributions. Blundell et al. (2008) use consumption responses to income shocks to measure the extent of household insurance. Following Sargent

et al. (2026a), we estimate the vector of log consumption at each of M percentile groups, $c_t = (c_{0,t}, \dots, c_{M-1,t})^\top$, defined in (46). We use linked household interviews to estimate the transition matrix Π governing movement among its indices.

Berger et al. (2023) also summarize CEX households with a few cross-sectional statistics. They show that a class of New Keynesian economies with incomplete markets has a representative-agent representation with two wedges, one on the discount factor and one on labor supply, that are functions of households' consumption shares and relative wages. They measure the wedges from CEX panel data for 1992–2017 and find that departures from perfect risk sharing account for about 7 percent of U.S. output volatility on average and about a third of the output loss in the Great Recession. Their wedges carry the aggregate implications of the cross-section. Our state vector carries its implications for percentile-specific continuation values.

Jappelli and Pistaferri (2006) demonstrate the value of consumption-transition matrices for understanding intertemporal behavior and consumption mobility. Richer empirical income processes emphasize nonlinear persistence, heterogeneous responses, and life-cycle dynamics, as in Guvenen et al. (2021) and Arellano et al. (2024). Our transition matrix determines the probabilities of the future positions over which a household's continuation value is evaluated. We allow households to doubt those probabilities.

Our treatment of model uncertainty follows the multiplier-preference and robust-control literature of Hansen and Sargent (2001, 2008b) and Barillas et al. (2009), in which a relative-entropy penalty regulates the concern for misspecification and connects worst-case valuation to the statistical distinguishability of competing models. Hansen and Sargent (2007) and Hansen and Sargent (2022) emphasize that misspecification can be structured by allowing different components of a model to be distrusted separately and disciplined by different penalties. Related recursive approaches to ambiguity and misspecification include Klibanoff et al. (2009) and Hansen and Miao (2018). We apply this idea to the joint law of the next-period percentile and consumption innovations, which the entropy chain rule factors into a nested recursion. Separate penalties let us distinguish the welfare effects of doubts about each component.

Strzalecki (2013) provides a complementary preference-theoretic perspective. In recursive models of ambiguity, the order in which uncertainty is aggregated carries behavioral content. Among standard ambiguity-averse specifications only maxmin expected utility is indifferent to the timing of the resolution of uncertainty, and in his discounted setting multiplier preferences prefer earlier resolution. Our recursion nests two sources of uncertainty within a single period transition. When the two penalties differ, distorting the next-period percentile first and the consumption innovation second is a different specification from the

reverse ordering. We condition first on the household's destination and then on consumption innovations there to match its two forecasting problems.

The rest of the paper is organized as follows. Section 2 lays out the environment, uses the entropy chain rule to derive a two-stage recursion with separate penalties on transition and innovation distortions, defines the four cases, and derives affine value functions for Cases 0 and 1. Section 3 develops the welfare accounting. It defines the compensating difference, gives closed-form welfare measures for Cases 0 and 1, and proves that without model uncertainty, mobility redistributes welfare costs across percentiles while leaving the long-run average cost unchanged. Section 4 takes the model to CEX data. It describes the samples and percentile assignments, the estimated transition matrices, the DMD state construction, and the detection-error calibration of the two penalties. Section 5 reports welfare costs across percentiles under both ranking schemes and decomposes them by source. Section 6 concludes. Appendices collect proofs, detailed derivations of Cases 1–3, computational details, and supplementary empirical results.

2 Two uncertainties

We study a household with multiplier preferences that doubts the conditional distribution of next-period consumption at a given percentile and the transition probabilities across percentiles. We describe our statistical model in Section 2.1 and introduce multiplier preferences in Section 2.2. Section 2.3 assigns separate relative entropy penalties to that conditional distribution and to perturbations to the transition matrix. We then define the four cases used in the empirical analysis (Section 2.4) and derive value functions for the two cases that have closed-form solutions (Section 2.5).

2.1 The environment

At date $t \in \mathbb{N}$ the state is (p_t, ζ_t, c_t) , where

- $p_t \in \{0, \dots, M-1\}$ is the household's percentile index,
- $\zeta_t \in \mathbb{R}^n$ is a state vector, and
- $c_t := (c_{0,t}, \dots, c_{M-1,t})^\top \in \mathbb{R}^M$ is the vector of logarithms of consumption at the M percentiles, where $c_{q,t}$ is the log consumption at percentile q . The household's log consumption at date t is therefore $c_{p_t,t}$.

We denote the state space by $\mathbf{S} := \{0, \dots, M-1\} \times \mathbb{R}^n \times \mathbb{R}^M$. A realization of $\{p_t\}_{t=0}^\infty$ describes the path of an infinitely lived household through percentiles of a cross-section.

2.1.1 State dynamics

Vectors ζ_t and c_t evolve according to

$$\zeta_{t+1} = A\zeta_t + C\varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I_k), \quad (1)$$

$$c_{q,t+1} - c_{q,t} = D_q\zeta_t + F_q\varepsilon_{t+1}, \quad q = 0, \dots, M-1, \quad (2)$$

where $A \in \mathbb{R}^{n \times n}$, $C \in \mathbb{R}^{n \times k}$, and, for each percentile q , $D_q \in \mathbb{R}^{1 \times n}$ and $F_q \in \mathbb{R}^{1 \times k}$ are the q -th rows of matrices $D \in \mathbb{R}^{M \times n}$ and $F \in \mathbb{R}^{M \times k}$.

The vector ζ_t predicts changes in c_t . Eigenvalues of A determine the persistence of the state vector ζ . The matrix C describes how the k -dimensional shocks ε_{t+1} affect ζ at time $t+1$.

Stacking (2) across q gives $c_{t+1} - c_t = D\zeta_t + F\varepsilon_{t+1}$. Consequently,

$$\mathbb{E}[c_{t+1} - c_t \mid \zeta_t, c_t] = D\zeta_t, \quad \text{Var}(c_{t+1} - c_t \mid \zeta_t, c_t) = FF^\top.$$

Row D_q maps ζ_t into the conditional mean of log-consumption growth at percentile q . The row F_q describes how the shocks ε_{t+1} affect $c_{q,t+1}$. Because all percentiles share the same innovations, F also determines the conditional comovement of consumption growth across percentiles.

The percentile state follows a Markov chain with transition matrix $\Pi \in \mathbb{R}^{M \times M}$:

$$\text{Prob}(p_{t+1} = j \mid p_t = p) = \Pi_{pj}, \quad \Pi_{pj} \geq 0, \quad \sum_{j=0}^{M-1} \Pi_{pj} = 1. \quad (3)$$

The shocks ε_{t+1} are common to the entire cross-section. They affect c_{t+1} through F and subsequent values of c_t through (A, C, D) , but they carry no information about dispersion among the households at a given percentile. Individual uncertainty enters through the transition matrix Π . Realized household consumption growth is $c_{p_{t+1},t+1} - c_{p_t,t}$, so both the vector c_t and the household's index p_t can change between dates. We estimate the law of motion of c_t from repeated cross-sections and Π from linked interviews. Consumption growth varies continuously among households within each percentile group. Our model represents the cross-section by M group means and omits dispersion within groups. This approximation allows estimation from a synthetic panel constructed from repeated CEX cross-sections and keeps the welfare analysis with two sources of model uncertainty tractable.

At state (p, ζ, c) , per-period utility is

$$(1 - \beta) e_p^\top c = (1 - \beta) c_p, \quad \beta \in (0, 1), \quad (4)$$

where c_p is the p -th element of the vector c and e_p is the p -th standard basis vector in $\mathbb{R}^M$. Since c_p is *log* consumption, (4) is logarithmic utility, and the factor $(1 - \beta)$ puts the value function in units of permanent log consumption.

Given the current state (p, ζ, c) , the approximating model specifies a joint conditional probability distribution over (p', ε) :

$$Q(\{j\} \times d\varepsilon \mid p, \zeta, c) = \Pi_{pj} \varphi(\varepsilon) d\varepsilon, \quad (5)$$

where φ denotes the $N(0, I_k)$ density. The next period state (ζ', c') follows from (ζ, c, ε) via (1)–(2). The two parts of Q , namely the transition probabilities and the innovation density, are the two components of the model that a household can distrust. Section 2.3 assigns each component its own relative entropy penalty.

2.1.2 Discussion of the specification

The specification is finite-dimensional, linear, Gaussian, and homoskedastic. These restrictions leave persistence in conditional means unconstrained while ruling out nonlinear dynamics and state-dependent conditional variances. With $M = 1$, the system contains the homoskedastic version of the persistent expected-growth component of Bansal and Yaron (2004), with ζ_t in the role of the small predictable state in consumption growth. With a predictable state and constant volatility, our specification is statistically closer to the Gaussian vector autoregressions with which Hansen et al. (2008) measure long-run risk.

A unit innovation ε_{t+1} changes log-consumption growth at percentile q contemporaneously by F_q and changes expected growth h periods later by $D_q A^{h-1} C$, for $h \geq 1$. Summing these impulse responses gives the innovation's long-run effect on the level of log consumption at percentile q ,

$$F_q + \sum_{r=0}^{\infty} D_q A^r C, \quad (6)$$

which is the loading of ε_{t+1} in the martingale component of the additive functional decomposition of Section 4.3. In the system that we estimate, this loading is the same at every percentile. The percentile series share one trend and one martingale and differ only by stationary components, so our model imposes that the permanent effect of an innovation is common to the entire cross-section. Because Section 4.3 models the demeaned percentile vectors as a stable vector autoregression, decomposition (53) holds as an identity, and no shock can permanently widen the distribution, even though inequality trended over our sample.

We call the matrix A *stable* if the eigenvalues of its nonconstant block lie inside the unit circle. The definition ignores the unit eigenvalue that a constant coordinate of ζ carries,

because that eigenvalue produces deterministic growth. If ζ has no constant coordinate, the nonconstant block is A itself. The sum in (6) converges whenever A is stable, the condition that also makes the discounted compensating differences of Section 3 finite (Appendix D). When the dominant eigenvalues of A lie close to unity, however, the terms of the sum decay slowly, and small innovations to the conditional mean accumulate into large revisions of the discounted consumption path even though the components responsible are hard to detect.

2.2 Multiplier preferences

Following Hansen and Sargent (2001, 2008b) and Barillas et al. (2009), we assume that the household regards the kernel Q as an approximation that it fears is misspecified. For probability measures $P \ll R$ we write $\text{ent}(P\|R) := \int \log \frac{dP}{dR} dP$ for relative entropy. Throughout, v_p denotes the value of a household at percentile p , and we assume that the exponential moments appearing below are finite.

2.2.1 The $\mathcal{T}_\theta$ operator

Fix the current state (p, ζ, c) and let $\mathcal{H}(p, \zeta, c)$ denote the set of all measurable likelihood ratios $h(p', \varepsilon) \geq 0$ satisfying

$$\mathbb{E}_Q [h(p', \varepsilon) | p, \zeta, c] = 1. \quad (7)$$

We use the lower-semicontinuous convention $0 \log 0 = 0$. The $\mathcal{T}_\theta$ operator applied to a discounted continuation value $\beta v(\cdot)$ is

$$\mathcal{T}_\theta[\beta v](p, \zeta, c) := \min_{h \in \mathcal{H}(p, \zeta, c)} \mathbb{E}_Q \left[h(p', \varepsilon) \left(\beta v_{p'}(\zeta', c') + \theta \log h(p', \varepsilon) \right) \middle| p, \zeta, c \right]. \quad (8)$$

This corresponds to the operator $\mathcal{T}_1$ defined by Hansen and Sargent (2008a). We seek a value function that satisfies the Bellman equation

$$v_p(\zeta, c) = (1 - \beta)c_p + \mathcal{T}_\theta[\beta v](p, \zeta, c). \quad (9)$$

Lemma 1. *Assume that $Y := \beta v_{p'}(\zeta', c')$ is finite $Q(\cdot | p, \zeta, c)$ -a.s. and that*

$$0 < \mathbb{E}_Q[\exp(-Y/\theta) | p, \zeta, c] < \infty.$$

The minimizer of

$$\min_{h \geq 0, \mathbb{E}_Q[h | p, \zeta, c] = 1} \mathbb{E}_Q \left[h \left(\beta v_{p'}(\zeta', c') + \theta \log h \right) \middle| p, \zeta, c \right] \quad (10)$$

is unique up to $Q(\cdot \mid p, \zeta, c)$ -a.s. equivalence and is given on the support of Q by

$$h^*(p', \varepsilon \mid p, \zeta, c) = \frac{\exp\left(-\frac{\beta}{\theta} v_{p'}(\zeta', c')\right)}{\mathbb{E}_Q\left[\exp\left(-\frac{\beta}{\theta} v_{p'}(\zeta', c')\right) \mid p, \zeta, c\right]}. \quad (11)$$

The minimized value is

$$\mathcal{T}_\theta[\beta v](p, \zeta, c) = -\theta \log \mathbb{E}_Q\left[\exp\left(-\frac{\beta}{\theta} v_{p'}(\zeta', c')\right) \mid p, \zeta, c\right]. \quad (12)$$

Proof. See Appendix A.1.

The worst-case kernel exponentially tilts the approximating kernel toward outcomes with low continuation values and converges back to the approximating kernel as $\theta \rightarrow \infty$.

2.3 Separate penalties for innovations and transitions

The approximating kernel (5) has two parts, the transition probabilities Π in (3) and the innovation density φ governing (1)–(2). A household may distrust these components to different degrees, having learned about each from different data. We represent these concerns with separate relative entropy penalties.

Informal observations and systematic empirical evidence both suggest that households have less precise knowledge of transition probabilities than of the innovation distribution. The reason is an asymmetry in what a household observes. The innovation ε_{t+1} is common to the entire cross-section, so a household that watches the distribution for T quarters sees T draws of it. Percentile transitions are individual. Over the same T quarters, the household observes only its own sequence of transitions. Learning about other transitions from other households is difficult. Estimates of mobility differ across data sets, sampling frequencies, and definitions of position (Jäntti and Jenkins, 2015), and much of the measured movement reflects transitory shocks and measurement error rather than persistent reranking (Bonhomme and Robin, 2009, Glewwe, 2012).

Our estimate of Π has an additional source of imprecision. A transition matrix over M positions has $M(M-1)$ free parameters, and the estimator (47) spreads the linked interviews across M^2 cells. The two arguments imply different distortions, because we estimate Π from many households at once while a household experiences only one history. Section 4.6 calibrates the penalties using the household's information.

At a fixed current state, an alternative kernel $\tilde{Q} \ll Q$ can be factored as

$$\tilde{Q}(\{j\} \times d\varepsilon \mid p, \zeta, c) = \tilde{\Pi}_{pj} \tilde{\varphi}_j(\varepsilon \mid \zeta, c) d\varepsilon, \quad (13)$$

where $\tilde{\Pi}_p$ is a probability vector on $\{0, \dots, M-1\}$ and each $\tilde{\varphi}_j(\cdot \mid \zeta, c)$ is a probability density. The chain rule for relative entropy gives

$$\text{ent}(\tilde{Q} \parallel Q) = \underbrace{\sum_{j=0}^{M-1} \tilde{\Pi}_{pj} \log \frac{\tilde{\Pi}_{pj}}{\Pi_{pj}}}_{=: \text{ent}(\tilde{\Pi}_p \parallel \Pi_p)} + \sum_{j=0}^{M-1} \tilde{\Pi}_{pj} \text{ent}(\tilde{\varphi}_j(\cdot \mid \zeta, c) \parallel \varphi). \quad (14)$$

Absolute continuity implies the support restriction $\tilde{\Pi}_{pj} = 0$ whenever $\Pi_{pj} = 0$.

Factorization (13) draws a next-period percentile first and an innovation conditional on that percentile. We use it to distinguish uncertainty about which percentile a household moves to next from uncertainty about the conditional distribution of (ζ', c') given $p' = j$.

The chain rule applies equally to the reverse ordering, in which the innovation is drawn first and the next-period percentile is conditional on it. When $\theta_\varepsilon = \theta_\Pi$, the two orderings give the same recursion, because the two entropy terms then sum to $\theta \text{ent}(\tilde{Q} \parallel Q)$ under either factorization. When the penalties differ, the ordering affects preferences (Strzalecki, 2013). In (14), we condition on the household's destination before considering consumption innovations there.

We attach penalty $\theta_\Pi > 0$ to the first term in factorization (14) and $\theta_\varepsilon > 0$ to the second. The household's continuation-value problem is

$$\min_{\tilde{\Pi}_p, \{\tilde{\varphi}_j\}} \sum_{j=0}^{M-1} \tilde{\Pi}_{pj} \left[\int \beta v_j(\zeta', c') \tilde{\varphi}_j(\varepsilon) d\varepsilon + \theta_\varepsilon \text{ent}(\tilde{\varphi}_j \parallel \varphi) \right] + \theta_\Pi \text{ent}(\tilde{\Pi}_p \parallel \Pi_p). \quad (15)$$

If $\theta_\varepsilon = \theta_\Pi = \theta$, (15) reduces to the single-penalty problem (10).

To solve (15), we first minimize over the conditional innovation densities. For each next-period percentile j , Lemma 1 gives the robust certainty equivalent

$$(\mathcal{R}_\varepsilon v_j)(\zeta, c) := -\theta_\varepsilon \log \mathbb{E}_\varphi \left[\exp \left(-\frac{\beta}{\theta_\varepsilon} v_j(\zeta', c') \right) \mid \zeta, c \right]. \quad (16)$$

The outer minimization over transition probabilities is itself an entropy-penalized problem on a finite set.

Proposition 2 (Two-stage recursion). *Suppose the exponential normalizer in (16) is finite*

and positive for every j in the support of Π_p . The value of the two-penalty problem (15) is

$$-\theta_\Pi \log \left(\sum_{j=0}^{M-1} \Pi_{pj} \exp \left(-\frac{(\mathcal{R}_\varepsilon v_j)(\zeta, c)}{\theta_\Pi} \right) \right). \quad (17)$$

Consequently, the household's Bellman equation associated with penalties $(\theta_\varepsilon, \theta_\Pi)$ is

$$v_p(\zeta, c) = (1 - \beta)c_p - \theta_\Pi \log \left(\sum_{j=0}^{M-1} \Pi_{pj} \exp \left(-\frac{(\mathcal{R}_\varepsilon v_j)(\zeta, c)}{\theta_\Pi} \right) \right). \quad (18)$$

Minimizers are the exponentially tilted transition probabilities

$$\tilde{\Pi}_{pj}(\zeta, c) = \frac{\Pi_{pj} \exp \left(-(\mathcal{R}_\varepsilon v_j)(\zeta, c)/\theta_\Pi \right)}{\sum_k \Pi_{pk} \exp \left(-(\mathcal{R}_\varepsilon v_k)(\zeta, c)/\theta_\Pi \right)} \quad (19)$$

and, conditional on next-period percentile j , the exponentially tilted density of consumption innovations

$$\tilde{\varphi}_j(\varepsilon \mid \zeta, c) = \varphi(\varepsilon) \frac{\exp \left(-(\beta/\theta_\varepsilon)v_j(\zeta', c') \right)}{\mathbb{E}_\varphi \left[\exp \left(-(\beta/\theta_\varepsilon)v_j(\zeta', c') \right) \mid \zeta, c \right]}. \quad (20)$$

Proof. See Appendix A.2.

The inner minimization distorts the innovation distribution conditional on the next-period percentile, and the outer minimization distorts the probabilities of those percentiles. Because the inner certainty equivalents enter the transition tilt (19), changing the innovation distribution can also change the worst-case transition probabilities.

We can expand the recursion for large θ_ε and θ_Π to identify uncertainties that enter at the two stages. Fix a value function v and a state (p, ζ, c) . The continuation-value aggregator is

$$\mathcal{T}_{\theta_\varepsilon, \theta_\Pi}[\beta v](p, \zeta, c) := -\theta_\Pi \log \left\{ \sum_{j=0}^{M-1} \Pi_{pj} \left(\mathbb{E}_\varphi \left[\exp \left(-\frac{\beta}{\theta_\varepsilon} v_j(\zeta', c') \right) \mid \zeta, c \right] \right)^{\theta_\varepsilon/\theta_\Pi} \right\}. \quad (21)$$

Define $Y_j := \beta v_j(\zeta', c')$. For each $j \in \text{supp}(\Pi_p)$, let $\mathcal{K}_j(t) := \log \mathbb{E}_\varepsilon[e^{tY_j}]$ be its conditional cumulant-generating function. We use $\mathbb{E}_\varepsilon$ and Var_ε for innovation moments and $\mathbb{E}_{\Pi_p}$ and Var_{Π_p} for moments over next-period percentiles $j \sim \Pi_p$.

Proposition 3. *Suppose that each $\mathcal{K}_j$ is finite on a neighborhood of zero. Then, as $\theta_\varepsilon \rightarrow \infty$ and $\theta_\Pi \rightarrow \infty$,*

$$\begin{aligned} \mathcal{T}_{\theta_\varepsilon, \theta_\Pi}[\beta v](p, \zeta, c) &= \mathbb{E}_{\Pi_p}[\mathbb{E}_\varepsilon[Y_j]] - \frac{1}{2\theta_\varepsilon} \mathbb{E}_{\Pi_p}[\text{Var}_\varepsilon(Y_j)] - \frac{1}{2\theta_\Pi} \text{Var}_{\Pi_p}(\mathbb{E}_\varepsilon[Y_j]) \\ &\quad + O(\theta_\varepsilon^{-2} + \theta_\Pi^{-2} + (\theta_\varepsilon \theta_\Pi)^{-1}), \end{aligned} \quad (22)$$

pointwise in (p, ζ, c) .

Proof. See Appendix A.3.

Equation (22) provides a first-order decomposition of a risk-sensitivity adjustment. The leading term is the expected continuation value under the approximating model. The first correction is the expected conditional variance of the continuation value generated by innovations,

$$\mathbf{D}_p := \mathbb{E}_{\Pi_p}[\text{Var}_\varepsilon(Y_j)]. \quad (23)$$

This is the exposure associated with uncertainty about the conditional distribution of innovations.

The second correction is the variance, across the percentiles the household may reach, of the conditional expected continuation value,

$$\mathbf{M}_p := \text{Var}_{\Pi_p}(\mathbb{E}_\varepsilon Y_j), \quad (24)$$

which measures exposure to uncertainty about Π_p . We collect both across percentiles as $\mathbf{D} := (\mathbf{D}_0, \dots, \mathbf{D}_{M-1})^\top$ and $\mathbf{M} := (\mathbf{M}_0, \dots, \mathbf{M}_{M-1})^\top$. We use these exposures to compare the two welfare costs in Section 5.6.

Remark 4 (The innovation term is exact in the linear–Gaussian model). Suppose v is affine, as in Proposition 5, and $\theta_\Pi = \infty$. Then (22) holds with no remainder. Conditional on the next-period percentile j , the quantity Y_j is Gaussian, so the inner expansion (75) is the exact Gaussian identity $-\theta \log \mathbb{E}[e^{-X/\theta}] = \mathbb{E}[X] - \text{Var}(X)/(2\theta)$. With $\theta_\Pi = \infty$, the outer stage is an expectation, which the expansion reproduces exactly. For affine v and finite θ_Π , the remainder in Proposition 3 comes entirely from the outer stage, which averages over a finite set of percentiles.

For an affine value function, $\text{Var}_\varepsilon(Y_j) = \beta^2 \mathbf{e}_j$, so $\mathbf{D} = \beta^2 \Pi \mathbf{e}$ and the propagated innovation term $(I - \beta \Pi)^{-1} \mathbf{D} / (2\theta_\varepsilon)$ equals the closed-form Case 1 increment $-\kappa$ of (36). Section 5.6 approximates the transition term alone.

2.4 Four cases

Setting each penalty either to a finite value or to infinity gives four combinations:

	No doubt on Π $\theta_\Pi = \infty$	Doubt on Π $\theta_\Pi < \infty$
No doubt on ε $\theta_\varepsilon = \infty$	Expected utility (Case 0)	Transition uncertainty (Case 2)
Doubt on ε $\theta_\varepsilon < \infty$	Innovation uncertainty (Case 1)	Joint uncertainty (Case 3)

Each case is covered by a specialization of Proposition 2.

In *Case 0*, $\theta_\varepsilon = \theta_\Pi = \infty$, distortions are infinitely expensive, $h \equiv 1$, and

$$v_p(\zeta, c) = (1 - \beta)c_p + \beta \sum_{j=0}^{M-1} \Pi_{pj} \mathbb{E}_\varphi [v_j(\zeta', c') | \zeta, c]. \quad (25)$$

In *Case 1*, the household is uncertain about the distribution of ε_{t+1} in (1)–(2). Here $\theta_\varepsilon = \theta$ and $\theta_\Pi = \infty$, so the conditional distribution of innovations may be distorted separately for each next-period percentile, but the transition probabilities remain those of the approximating model:

$$v_p(\zeta, c) = (1 - \beta)c_p - \theta \sum_{j=0}^{M-1} \Pi_{pj} \log \mathbb{E}_\varphi \left[\exp \left(-\frac{\beta}{\theta} v_j(\zeta', c') \right) \middle| \zeta, c \right], \quad (26)$$

with the worst-case innovation likelihood ratio, conditional on landing at j ,

$$\frac{d\tilde{\varphi}_j}{d\varphi}(\varepsilon | \zeta, c) =: g_j^*(\varepsilon | \zeta, c) = \frac{\exp \left(-\frac{\beta}{\theta} v_j(\zeta', c') \right)}{\mathbb{E}_\varphi \left[\exp \left(-\frac{\beta}{\theta} v_j(\zeta', c') \right) \middle| \zeta, c \right]}. \quad (27)$$

The corresponding Lebesgue density is $\tilde{\varphi}_j(\varepsilon | \zeta, c) = \varphi(\varepsilon)g_j^*(\varepsilon | \zeta, c)$, and $\mathbb{E}_\varphi[g_j^* | \zeta, c] = 1$ separately for every next-period percentile j . The separate normalizations preserve the transition probabilities Π_p . Since the conditional innovation density varies with j , the alternative law generally makes innovations and next-period percentiles dependent. (See Appendix B.1 for a detailed derivation.)

In *Case 2*, the household is uncertain about the transition probabilities across percentiles. Here $\theta_\varepsilon = \infty$ and $\theta_\Pi = \theta$, so the transition probabilities may be distorted but the distribution of consumption innovations remains that of the approximating model. Writing

$$\bar{v}_j(\zeta, c) := \mathbb{E}_\varphi [v_j(\zeta', c') | \zeta, c] \quad (28)$$

gives

$$v_p(\zeta, c) = (1 - \beta)c_p - \theta \log \left(\sum_{j=0}^{M-1} \Pi_{pj} \exp \left(-\frac{\beta}{\theta} \bar{v}_j(\zeta, c) \right) \right), \quad (29)$$

with worst-case transition probabilities

$$\tilde{\Pi}_{pj}(\zeta, c) = \frac{\Pi_{pj} \exp\left(-\frac{\beta}{\theta} \bar{v}_j(\zeta, c)\right)}{\sum_{k=0}^{M-1} \Pi_{pk} \exp\left(-\frac{\beta}{\theta} \bar{v}_k(\zeta, c)\right)}. \quad (30)$$

Because the likelihood ratio does not depend on ε , innovations retain their $N(0, I_k)$ distribution. (See Appendix B.2 for a detailed derivation.)

In *Case 3* (joint uncertainty), $\theta_\varepsilon = \theta_\Pi = \theta$, and any nonnegative likelihood ratio h satisfying the mean-one constraint is admissible. With

$$\mathcal{M}_j(\zeta, c) := \mathbb{E}_\varphi \left[\exp\left(-\frac{\beta}{\theta} v_j(\zeta', c')\right) \middle| \zeta, c \right], \quad (31)$$

Lemma 1 gives

$$v_p(\zeta, c) = (1 - \beta)c_p - \theta \log \left(\sum_{j=0}^{M-1} \Pi_{pj} \mathcal{M}_j(\zeta, c) \right). \quad (32)$$

(See Appendix B.3 for a detailed derivation.)

With equal penalties, Case 3 is a single relative-entropy distortion of the joint kernel. Its Bellman equation is equivalent to an Epstein–Zin recursion with intertemporal elasticity of substitution equal to one and relative risk aversion $1 + \beta/\theta$. Remark 10 in Appendix C shows this equivalence.

2.5 Closed-form solutions for Cases 0 and 1

In the linear–Gaussian model of Section 2, doubts about the innovation distribution lower the value function through an intercept adjustment while leaving the loadings on current consumption and the predictors unchanged.

Proposition 5. *Assume that $I_{Mn} - \beta(A^\top \otimes \Pi)$ is nonsingular and that the conditions of Proposition 13 hold. We normalize the Case 0 intercept to zero.⁴ The Case 0 and Case 1 value functions have common slope vectors λ_p and b_p :*

$$v_p^{(0)}(\zeta, c) = \lambda_p^\top \zeta + b_p^\top c, \quad v_p^{(1)}(\zeta, c) = v_p^{(0)}(\zeta, c) + \kappa_p. \quad (33)$$

Let $W := [b_0, \dots, b_{M-1}]^\top$ and $\Lambda := [\lambda_0, \dots, \lambda_{M-1}]^\top$. These matrices are the unique solution

⁴If ζ contains a constant coordinate, its loading and a separate intercept are not jointly identified. We use the same normalization for its loading in Cases 0 and 1 and assign the constant difference between the two value functions to κ . If ζ has no constant coordinate, the Case 0 Bellman equation implies a zero intercept.

to

$$W = (1 - \beta)(I - \beta\Pi)^{-1}, \quad (34)$$

$$\Lambda = \beta\Pi\Lambda A + \beta\Pi W D. \quad (35)$$

For $s_j := C^\top \lambda_j + F^\top b_j$, $\mathbf{e}_j := \|s_j\|^2$, and $\mathbf{e} := (\mathbf{e}_0, \dots, \mathbf{e}_{M-1})^\top$, the Case 1 intercept is

$$\kappa = -\frac{\beta^2}{2\theta}(I - \beta\Pi)^{-1}\Pi\mathbf{e}, \quad (36)$$

and, conditional on next-period percentile j , its worst-case innovation distribution is

$$\varepsilon \sim N(-(\beta/\theta)s_j, I_k).$$

In particular, $\kappa_p \leq 0$ for every p .

Proof. See Appendix A.4.

The entries of b_p are non-negative and sum to one. They weight c by discounted future occupancy probabilities. Conditional on reaching percentile j , $v_j(\zeta', c')$ is Gaussian with variance $\mathbf{e}_j$, so its entropy adjustment $-\beta^2\mathbf{e}_j/(2\theta)$ is constant in (ζ, c) and leaves both slope vectors unchanged. The worst-case innovation mean shifts to $-(\beta/\theta)s_j$, and κ_p collects the discounted value effects.

When $M = 1$, (36) reduces to the scalar multiplier-preference formula of Hansen and Sargent (2008a). The Markov resolvent $(I - \beta\Pi)^{-1}$ in (36) replaces the scalar resolvent $(1 - \beta)^{-1}$. Remark 8 in Appendix C gives the derivation.

In Cases 2 and 3, the log-sum-exponential terms in (29) and (32) generally produce nonaffine value functions, which we compute by value-function iteration. Proposition 13 in Appendix E establishes existence and uniqueness in a linear-growth class and convergence of value-function iteration, and Appendix H.1 describes the finite-grid implementation and establishes its convergence.

3 Welfare accounting

We convert value losses into units of consumption. Section 3.1 defines a compensating difference for each of the four cases of Section 2.4. Section 3.1.1 uses the value functions of Section 2.5 to derive closed forms for Cases 0 and 1 and states how mobility across percentiles affects the Case 0 welfare cost. Section 3.1.2 defines the numerical analogues for Cases 2 and 3.

3.1 Compensating differences

To express the welfare effects in units of consumption, we compare each case with a sequence that preserves conditional expected consumption in levels at every horizon. For each horizon and future percentile, let $\mu_{j,h}(\zeta_t, c_t)$ and $\sigma_{j,h}^2$ denote the conditional mean and variance of $c_{j,t+h}$. Iterating (1)–(2) gives $\mu_{j,h} = c_{j,t} + \psi_{j,h}^\top \zeta_t$ and $\sigma_{j,h}^2 = \sum_{i=0}^{h-1} \|J_{j,i}\|^2$, where

$$\psi_{j,h}^\top := \sum_{s=0}^{h-1} D_j A^s, \quad J_{j,i} := F_j + \sum_{r=0}^{i-1} D_j A^r C$$

accumulate the predictable-growth and innovation responses in the state dynamics. Appendix D gives the derivation. Following Obstfeld (1994) and Tallarini (2000), we define

$$\bar{c}_{j,h}(\zeta_t, c_t) := \log \mathbb{E}_\varphi[\exp(c_{j,t+h}) \mid \zeta_t, c_t] = \mu_{j,h}(\zeta_t, c_t) + \frac{1}{2}\sigma_{j,h}^2, \quad h \geq 0. \quad (37)$$

For a household currently at percentile p , we define the benchmark value w_p as normalized expected discounted log utility when log consumption at every horizon h and percentile j is the nonrandom amount $\bar{c}_{j,h}$, while the percentile itself still moves according to Π :

$$w_p(\zeta, c) = (1 - \beta) \sum_{h=0}^{\infty} \beta^h \sum_{j=0}^{M-1} [\Pi^h]_{pj} \bar{c}_{j,h}(\zeta, c). \quad (38)$$

The benchmark in (38) preserves mobility while removing consumption innovations. We use the same w_p in all four cases, so the four compensating differences share one benchmark and can be compared directly. In Cases 2 and 3, the worst-case distortion of the transition probabilities appears only in $v_p^{(m)}$, so its welfare cost is part of $\alpha_p^{(m)}$.

Adding a scalar a to every component of c shifts consumption permanently by a log points and gives

$$v_p^{(m)}(\zeta, c + a\mathbf{1}) = v_p^{(m)}(\zeta, c) + a, \quad w_p(\zeta, c + a\mathbf{1}) = w_p(\zeta, c) + a. \quad (39)$$

Appendix A.5 proves this translation property. The compensating difference $\alpha^{(m)}(\zeta, c) \in \mathbb{R}^M$, where $m \in \{0, 1, 2, 3\}$ indexes the four cases, is the permanent common increase in log consumption under Case m that equates its value to w_p :

$$v_p^{(m)}(\zeta, c + \alpha_p^{(m)}(\zeta, c)\mathbf{1}) = w_p(\zeta, c), \quad p = 0, \dots, M-1. \quad (40)$$

Equation (39) gives

$$\alpha_p^{(m)}(\zeta, c) = w_p(\zeta, c) - v_p^{(m)}(\zeta, c). \quad (41)$$

3.1.1 Closed-form compensating differences for Cases 0 and 1

Let $\sigma_h^2 := (\sigma_{0,h}^2, \dots, \sigma_{M-1,h}^2)^\top$ collect the horizon- h conditional variances across percentiles. The compensating differences below discount this sequence over horizons. The discounted variance series is finite when A is stable as defined in Section 2.1.2. Appendix D proves this and expresses stability of the matrices that we estimate in Section 4.5 as $\rho(\widehat{B}) < 1$ for the fitted full-state system and $\rho(\widetilde{B}) < 1$ for the reduced representation.

Proposition 6. *Assume that A is stable. The compensating differences for Cases 0 and 1 are*

$$\alpha^{(0)} = \frac{1-\beta}{2} \sum_{h=1}^{\infty} \beta^h \Pi^h \sigma_h^2, \quad (42)$$

$$\alpha^{(1)} = \alpha^{(0)} - \kappa = \alpha^{(0)} + \frac{\beta^2}{2\theta} (I - \beta\Pi)^{-1} \Pi \mathbf{e}. \quad (43)$$

Thus $\alpha_p^{(1)} \geq \alpha_p^{(0)}$ for every p .

Proof. See Appendix A.6.

Equation (42) is the welfare cost of consumption innovations under expected utility, measured relative to the conditional-mean consumption sequence. Case 1 adds the non-negative term $-\kappa$, the additional welfare cost from doubting the innovation distribution. Here $\mathbf{e}_j$ is the squared loading of the continuation value on the innovation, conditional on reaching percentile j .

Two special cases help us cross-check these formulas. If we shut down mobility by setting $\Pi = I$, each household stays at its own percentile and $\alpha_p^{(0)}$ collapses to $(\beta/2) \sum_{i \geq 0} \beta^i \|J_{p,i}\|^2$. Specializing further to a scalar consumption random walk, we get $\beta\sigma_\varepsilon^2/[2(1-\beta)]$ for Case 0 and $\alpha_p^{(1)}/\alpha_p^{(0)} = 1 + \beta/\theta_\varepsilon$ for the ratio of the two cases. These are the Obstfeld–Tallarini welfare cost and its linear scaling in risk aversion. Remark 11 in Appendix C records the derivations.

Equal exposures tell us when mobility matters for Case 1. Suppose every percentile carries the same exposure, $\mathbf{e}_j = \bar{\mathbf{e}}$. Then $\alpha^{(1)} - \alpha^{(0)} = \beta^2 \bar{\mathbf{e}} \mathbf{1}/[2\theta(1-\beta)]$, because $(I - \beta\Pi)^{-1} \Pi \mathbf{1} = (1-\beta)^{-1} \mathbf{1}$ for any stochastic Π . At this common exposure, doubting the innovation distribution costs the same at every percentile for any Π . The exposure itself depends on Π through the continuation-value loadings (Remark 9). Differences in exposure

allow mobility to redistribute the Case 1 increment across percentiles. We find in Section 5.4 that they differ very little, and the Case 1 increment is correspondingly flat.

Proposition 6 has the following consequence for the effect of mobility on the stationary-weighted mean.

Corollary 7. *Let π be a stationary distribution of Π , so that $\pi^\top \Pi = \pi^\top$. Then*

$$\sum_{p=0}^{M-1} \pi_p \alpha_p^{(0)}(\Pi) = \frac{1-\beta}{2} \sum_{h=1}^{\infty} \beta^h \pi^\top \sigma_h^2 = \sum_{p=0}^{M-1} \pi_p \alpha_p^{(0)}(I). \quad (44)$$

Proof. See Appendix A.7.

For the estimated transition matrices, π_p is the long-run fraction of time a household spends at percentile p (Section 4.2). These time shares are the weights in Corollary 7.

3.1.2 Cases 2 and 3

In Cases 2 and 3, the transition certainty equivalent is a log-sum-exponential of continuation values across percentiles. Because the differences among those values vary with the current state, the worst-case transition probabilities and the compensating differences are state dependent. We therefore solve the Case 2 and Case 3 Bellman equations and compute their compensating differences numerically. Section 4.7 defines the stationary-state average used to report them.

Holding the transition penalty θ_Π fixed across Cases 2 and 3 lets us order their welfare costs. Their Bellman operators then differ only in the inner stage. Case 2 uses the expectation $\beta \bar{v}_j(\zeta, c)$, and Case 3 uses the robust certainty equivalent $(\mathcal{R}_\varepsilon v_j)(\zeta, c)$, which Jensen's inequality places weakly below it at every next-period percentile j . The outer stage of (18) is increasing in its arguments, so the Case 3 operator lies below the Case 2 operator pointwise. Both operators are monotone (Appendix E), so Proposition 13 carries the ordering to the fixed points and gives $v_p^{(3)} \leq v_p^{(2)}$ at every state. Equivalently,

$$\alpha_p^{(3)}(\zeta, c) - \alpha_p^{(2)}(\zeta, c) = v_p^{(2)}(\zeta, c) - v_p^{(3)}(\zeta, c) \geq 0, \quad (45)$$

so adding doubt about innovations to doubt about transitions cannot reduce the welfare cost.

4 Estimation

We now estimate the approximating model from Consumer Expenditure Survey (CEX) microdata, calibrate $(\theta_\varepsilon, \theta_\Pi)$ using detection-error probabilities, and compute compensating differences for the four cases. The percentile series in Section 4.1 provide the consumption vector c_t and the other observable inputs to the state, while we use the linked-household transitions in Section 4.2 to estimate the transition matrix Π . Sections 4.3 and 4.5 describe how we construct the state and estimate the consumption dynamics (A, C, D, F) in (1)–(2). Section 4.4 presents descriptive percentile series from the common income-complete sample. Appendix F explains why the consumption-only specification uses a broader positive-consumption sample. Section 4.6 calibrates the robustness penalties $(\theta_\varepsilon, \theta_\Pi)$, and Section 4.7 solves for the four value functions and their compensating differences. We repeat this construction under consumption and private-income rankings because the ranking variable changes both the observable state and Π .

4.1 Data and percentile assignments

We use the public-use CEX Interview Survey from 1990Q1 through 2024Q1. Consecutive quarterly interviews for the same consumer unit provide the short panels we use to estimate mobility. From each interview we construct private income, post-tax income, and total consumption expenditure. Appendix F documents the construction and screening rules.

At each date we divide the full survey cross-section into $M = 100$ survey-weighted percentile groups, and the percentile series defined below are averages within those groups. The transition matrix instead uses households observed in consecutive quarters and assigns their percentiles within that linked sample. We consider two ranking variables:

- **Consumption ranking.** We assign households to percentile groups by total consumption expenditure. The observable state vector uses only c_t , yielding a state of dimension $1 + M$.
- **Private-income ranking.** We assign households to percentile groups by private income. The observable state vector stacks the percentile vectors of private income m_t , post-tax income d_t , and consumption c_t , each defined in (46) below, yielding a state of dimension $1 + 3M$.⁵

The two specifications use different samples because the consumption-only state does not require valid income observations. The private-income specification uses the income-complete sample. The consumption-ranked DMD state and transition matrix use the broader sample

⁵This differs from Sargent et al. (2026a), where each variable is ranked independently.

of observations with valid positive consumption. For a direct comparison of private income, post-tax income, and consumption, Figure 2 uses the income-complete sample under both rankings. Appendix F states the screens for each sample.

For either ranking, let ℓ index the individual consumer units we observe in the survey, and let $\mathcal{B}_{p,t}$ be the set of records assigned to percentile group p at date t . Let $\mathbf{w}_{\ell,t}$ denote FINLWT21, the final CEX survey weight measuring how many consumer units in the population record ℓ represents, and let $U_{\ell,t}^{(m)}$, $U_{\ell,t}^{(d)}$, and $U_{\ell,t}^{(c)}$ denote private income, post-tax income, and consumption in levels for record ℓ , respectively. For $u \in \{m, d, c\}$, we set

$$u_{p,t} := \log \left(\frac{\sum_{\ell \in \mathcal{B}_{p,t}} \mathbf{w}_{\ell,t} U_{\ell,t}^{(u)}}{\sum_{\ell \in \mathcal{B}_{p,t}} \mathbf{w}_{\ell,t}} \right). \quad (46)$$

In our model, a household at percentile p consumes the mean consumption of the households assigned to that percentile group. The logarithm of this mean is $c_{p,t}$ in (2).

The number of groups matters for the transition channel. Doubts about mobility operate through $\mathbf{M}_p$, the dispersion of expected continuation values across the destinations a household can reach (Section 2.3). Coarsening the partition averages consumption over more households and moves dispersion from between groups to within them, where it is unmodeled. It also changes the estimated transition matrix and innovation exposures. Assessing how the relative welfare costs of the two doubts depend on M therefore requires re-estimating the model for each partition. The split of Case 0 into its i.i.d. and persistent components may also change, because $\mathbf{f}_j = \|F_j\|^2$ is the one-quarter innovation variance of a group mean. We use $M = 100$ percentile groups.

4.2 Transition matrix

We estimate the transition matrix in (3) using consumer units observed in two consecutive quarterly cross-sections. Unit ℓ is interviewed at date t and the same unit is interviewed again at date $t + 1$, with its interview wave and calendar quarter both advancing by one. Since the CEX interviews a consumer unit at most four times, each unit supplies at most three such pairs. We keep a pair only when both interviews pass the screens for the relevant ranking specification in Appendix F, and we write $\mathcal{P}$ for the set of origin interviews (ℓ, t) that survive.

Let G_ℓ^t denote household ℓ 's percentile at date t . We rank these households within the linked sample at each date, giving G_ℓ^t and G_ℓ^{t+1} . The percentile groups in Section 4.1 use the full survey cross-section. The model uses the same percentile index p_t for mobility governed by Π and consumption $c_{p_t,t}$. We therefore assume that a rank in the linked sample

represents the same position in the distribution as the corresponding rank in the full cross-section. We then estimate the transition matrix as the weighted relative frequency of moves from percentile p at date t to percentile q at date $t+1$:

$$\hat{\Pi}_{pq} = \frac{\sum_{(\ell,t) \in \mathcal{P}} \mathbf{w}_{\ell t} \mathbb{1}\{G_{\ell}^t = p\} \mathbb{1}\{G_{\ell}^{t+1} = q\}}{\sum_{(\ell,t) \in \mathcal{P}} \mathbf{w}_{\ell t} \mathbb{1}\{G_{\ell}^t = p\}}, \quad p, q = 0, \dots, M-1, \quad (47)$$

where $\mathbf{w}_{\ell t}$ is the origin interview's FINLWT21 survey weight.

The rows of the estimated transition matrix sum to one by construction of (47). With cross-sectional CEX weights, the column sums depart slightly from one.⁶ Both estimated matrices are irreducible and aperiodic, so each has a unique stationary distribution π and Corollary 7 applies. The weight π_p is the long-run fraction of time that one household spends at percentile p under the estimated matrix. We use these weights to average welfare costs across percentiles.

Pooling the pairs assumes that transition probabilities are constant over time and across consumer units and do not depend on age, cohort, household composition, the continuous state, or aggregate conditions. The worst-case probabilities in (19) depend on the state through continuation values.

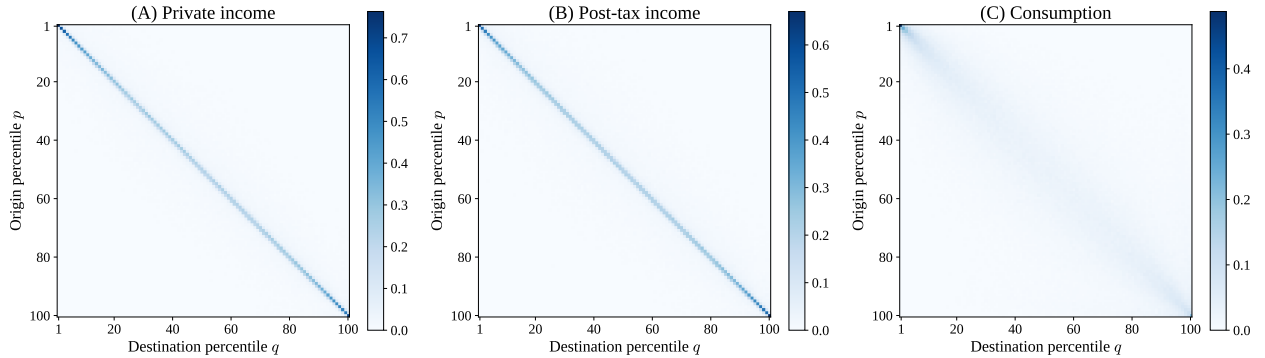


Figure 1: Estimated transition matrices $\hat{\Pi}_{pq}$ from the CEX panel with $M = 100$ percentiles. Left: private-income ranking. Center: post-tax-income ranking. Right: consumption ranking.

Figure 1 displays the estimated matrices for three alternative rankings, by private income, post-tax income, and consumption, each with $M = 100$ percentiles. All three exhibit strong diagonal persistence, but the off-diagonal structure differs markedly across rankings. The consumption matrix (right panel) is less concentrated around the diagonal than the

⁶Column sums range from 0.9866 to 1.0144 under private-income ranking and from 0.9921 to 1.0083 under consumption ranking. The total-variation distance between the stationary distribution and the uniform distribution is 0.0053 under private-income ranking and 0.0019 under consumption ranking.

private-income matrix (left panel), reflecting substantially more mobility across consumption percentiles than across private-income percentiles. The shares of estimated zero cells are 8.33% for private-income ranking, 5.96% for post-tax-income ranking, and 2.27% for consumption ranking. These cells remain zero under the worst-case distortion.

Transitory purchases, reporting error, and sampling variation all reshuffle households across consumption percentiles (Aguiar and Bils, 2015, Meyer and Sullivan, 2023). Because these data cannot separate persistent reranking from transitory variation, we report results under both rankings.

We report welfare estimates without standard errors, treating $\hat{\Pi}$ and the estimated consumption dynamics as fixed when solving the household's problem.

4.3 Additive functional and state construction

For each variable $u \in \{m, d, c\}$ (private income, post-tax income, consumption), collect the quantities defined in (46) into $u_t = (u_{0,t}, \dots, u_{M-1,t})^\top$. Let $r \in \{c, m\}$ denote the ranking variable. We define $Y_t := M^{-1} \sum_{p=0}^{M-1} r_{p,t}$ and the vectors relative to this index by $\tilde{u}_t := u_t - Y_t \mathbf{1}_M$. Thus, ΔY_t below is the change in the average log level of the ranking variable.

The choice of ranking variable determines the composition of the observable state vector.

Under **consumption ranking**, the state stacks growth of the consumption location index with the demeaned vector $\tilde{c}_t$:

$$y_t^{(c)} := \begin{pmatrix} \Delta Y_t - \nu \\ \tilde{c}_t - \boldsymbol{\nu}^{(c)} \end{pmatrix} \in \mathbb{R}^{1+M}, \quad (48)$$

where ν is the scalar trend in the location index and $\boldsymbol{\nu}^{(c)}$ is the time-series mean of the vector $\tilde{c}_t$. This $(1+M)$ -dimensional state captures dispersion across consumption percentiles and the dynamics of their location index. It is a direct generalization of the representative-agent ($M = 1$) case studied by Obstfeld (1994) and Tallarini (2000), where the state has one effective dimension and is given by a single consumption-growth series.

Under **private-income ranking**, following Sargent et al. (2026a), we stack the three vectors $\tilde{m}_t$, $\tilde{d}_t$, and $\tilde{c}_t$ with growth of the private-income location index:

$$y_t^{(m)} := \begin{pmatrix} \Delta Y_t - \nu \\ \tilde{m}_t - \boldsymbol{\nu}^{(m)} \\ \tilde{d}_t - \boldsymbol{\nu}^{(d)} \\ \tilde{c}_t - \boldsymbol{\nu}^{(c)} \end{pmatrix} \in \mathbb{R}^{1+3M}. \quad (49)$$

The richer $(1+3M)$ -dimensional state captures the joint dynamics of the private-income, post-tax-income, and consumption vectors, allowing the reduced-rank estimator to exploit covariation across all three variables.

In both cases we model the demeaned state as a first-order vector autoregression

$$y_{t+1} = B y_t + a_{t+1}, \quad \mathbb{E}[a_{t+1} \mid y_t] = 0, \quad \mathbb{E}[a_{t+1} a_{t+1}^\top] = \Omega, \quad (50)$$

following Sargent et al. (2026a). We use the VAR to separate the level of the location index into permanent and temporary parts. Proposition 1 of Sargent et al. (2026a) gives

$$Y_t = \nu t + \sum_{j=1}^t H^{\text{AF}} a_j - \mathbf{g} y_t + \mathbf{g} y_0 + Y_0, \quad (51)$$

where

$$\mathbf{g} = e_1^\top B(I - B)^{-1} = e_1^\top \sum_{h=1}^{\infty} B^h, \quad H^{\text{AF}} = e_1^\top + \mathbf{g} = e_1^\top (I - B)^{-1}. \quad (52)$$

Here e_1 is the first standard basis vector of the same dimension as y_t , so that $e_1^\top y_t$ is the $\Delta Y_t - \nu$ coordinate. The term νt is the deterministic trend in Y_t . An innovation a_j affects Y_j through $e_1^\top a_j$ and expected future growth through the VAR. $H^{\text{AF}} = e_1^\top (I - B)^{-1}$ combines these effects, so $\sum_{j=1}^t H^{\text{AF}} a_j$ is a martingale recording the permanent effect of past innovations. The term $-\mathbf{g} y_t$ is stationary when B is stable, and $\mathbf{g} y_0 + Y_0$ fixes the level at $t = 0$.

Corollary 1 of Sargent et al. (2026a) extends this result to every percentile and each variable $u \in \{m, d, c\}$. We can write

$$u_{p,t} = \delta_{u,p} + \nu t + \sum_{j=1}^t H^{\text{AF}} a_j + \chi_{u,p,t}, \quad (53)$$

where $\delta_{u,p}$ is a percentile-specific constant and $\chi_{u,p,t}$ is stationary. All percentile series share one trend and one martingale. They differ by the constants $\delta_{u,p}$ and the stationary terms $\chi_{u,p,t}$.

Decomposition (53) describes levels and long-run effects. A household's dynamic programming problem also requires a one-period state law, which we obtain by mapping the VAR directly into the state-space system (1)–(2). Under our Gaussian approximating model we choose a possibly rectangular matrix Γ with $\Gamma \Gamma^\top = \Omega$ and write $a_{t+1} = \Gamma \eta_{t+1}$, where $\eta_{t+1} \sim N(0, I_k)$. Appending a constant to the VAR state gives the predictor vector and

matrices in (1),

$$\zeta_t := \begin{bmatrix} 1 \\ y_t \end{bmatrix}, \quad A := \begin{bmatrix} 1 & 0 \\ 0 & B \end{bmatrix}, \quad C := \begin{bmatrix} 0 \\ \Gamma \end{bmatrix}, \quad (54)$$

since $\zeta_{t+1} = A\zeta_t + C\eta_{t+1}$ is then a restatement of (50), with η_{t+1} in the role that ε_{t+1} plays in (1). The first coordinate is constant, and the eigenvalues of B determine the persistence of y_t . The matrix Γ describes how the standard normal shocks η_{t+1} affect y_{t+1} , with $\Gamma\Gamma^\top = \Omega$.

We next derive the consumption-growth law. Let $e_1^\top$ select the first coordinate $\Delta Y_t - \nu$ from y_t , and let $e_q^{(c)}$ select the demeaned consumption coordinate for percentile q . Then $c_{q,t} = Y_t + \nu_q^{(c)} + e_q^{(c)} y_t$, so the stationary term in (53) is explicit for consumption:

$$c_{q,t} = \delta_q + \nu t + \sum_{s=1}^t H^{\text{AF}} a_s - (\mathbf{g} - e_q^{(c)}) y_t, \quad \delta_q := \mathbf{g} y_0 + Y_0 + \nu_q^{(c)}. \quad (55)$$

Taking first differences, substituting (50), and using $\mathbf{g}(I - B) = e_1^\top B$ gives

$$c_{q,t+1} - c_{q,t} = D_q \zeta_t + F_q \eta_{t+1}, \quad (56)$$

$$D_q := \begin{bmatrix} \nu & e_1^\top B + e_q^{(c)}(B - I) \end{bmatrix}, \quad (57)$$

$$F_q := (e_1^\top + e_q^{(c)})\Gamma. \quad (58)$$

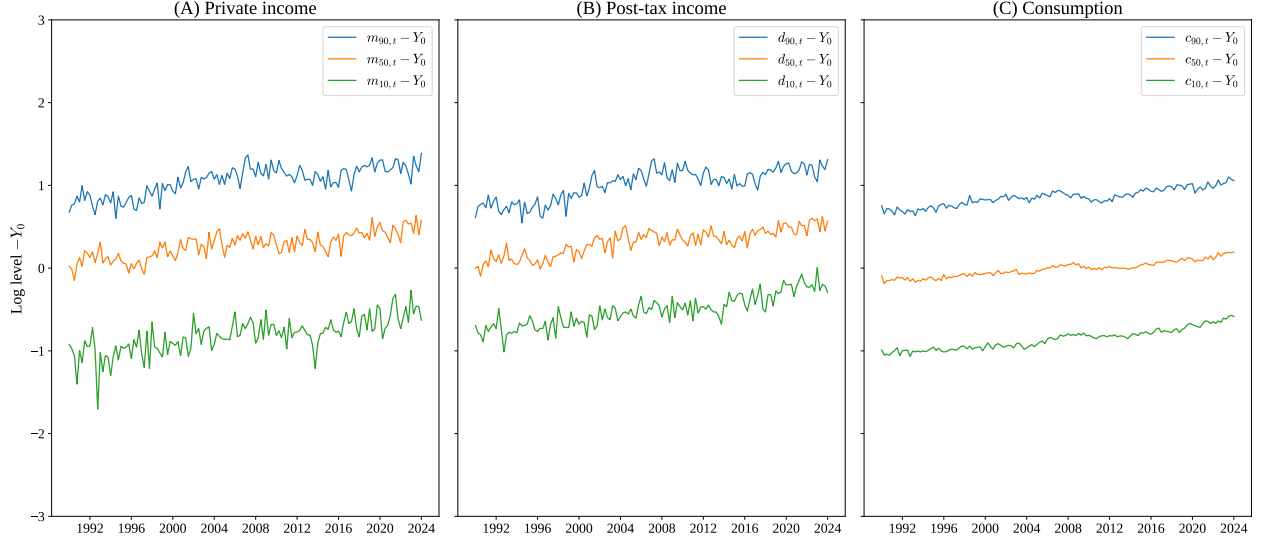
The two blocks of D_q match the two blocks of ζ_t . The drift ν multiplies the constant, and the remaining block multiplies y_t . Stacking D_q and F_q across percentiles gives D and F , so (1)–(2) now hold before dimension reduction with the matrices just defined.

4.4 Percentile series under the two rankings

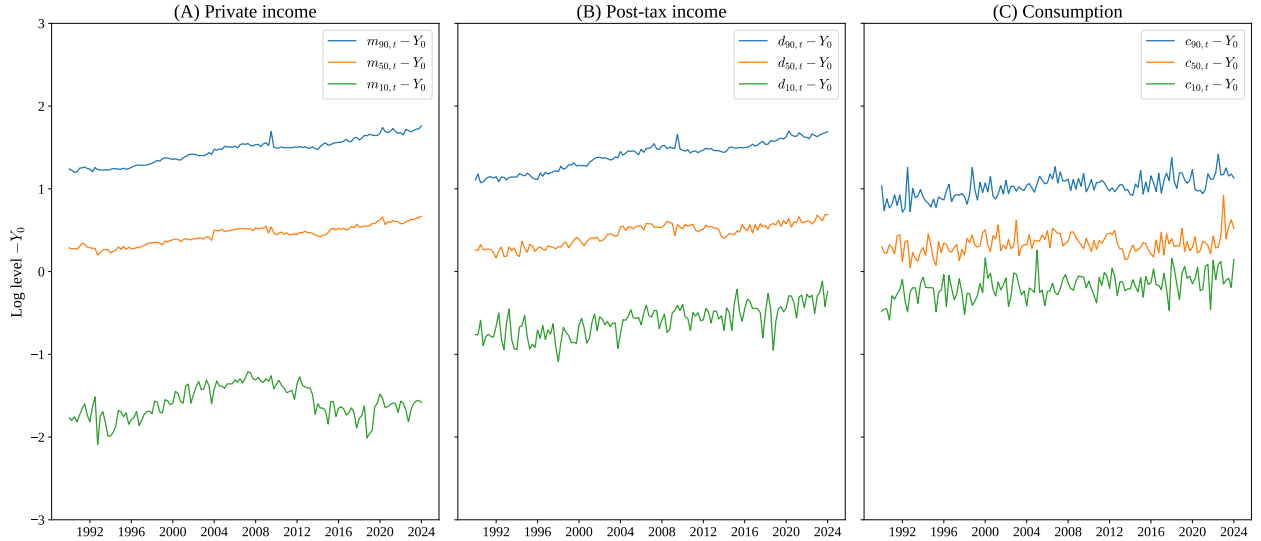
Figure 2 plots log levels of private income, post-tax income, and consumption at the 10th, 50th, and 90th percentiles relative to Y_0 under each ranking. Percentile means of the ranking variable move slowly. The means of the other variables fluctuate more from quarter to quarter because each percentile group combines households of different types. The two rankings also differ in scale. Under consumption ranking the observable state (48) has dimension $1 + M = 101$ and the additive functional decomposition of Y_t yields a trend growth rate $\nu = 0.0025$ per quarter, about one annual percentage point, while under private-income ranking the observable state (49) has dimension $1 + 3M = 301$.

Panel (a) sorts by consumption. The 10th–90th percentile spread in consumption is about 1.7 log points, and the plotted consumption series at those percentiles are slightly more volatile at the top and bottom of the distribution.

Panel (b) sorts by private income. The spread between the 10th and 90th percentiles is large in private income, about 3 log points, and the 10th private-income percentile displays substantial low-frequency movement in the series $m_{10,t}$, which rises into the late 2000s and falls through the 2010s. The three displayed consumption series in its subpanel C are more volatile over time than under consumption ranking, because private-income percentile groups mix households with heterogeneous consumption levels. In the plotted means, consumption exceeds post-tax income at the bottom and is below it at the median and top. The pattern is consistent with the lowest private-income groups holding a substantial share of retired and temporarily low-income households that dissave, and with working-age households saving a portion of income at the median and top.



(a) Consumption ranking. Log levels relative to Y_0 at the 10th, 50th, and 90th consumption percentiles: (A) private income, (B) post-tax income, and (C) consumption.



(b) Private-income ranking. Log levels relative to Y_0 at the 10th, 50th, and 90th private-income percentiles: (A) private income, (B) post-tax income, and (C) consumption.

Figure 2: Percentile series $u_{p,t}$ at selected values of p under the two ranking schemes. All panels use the common income-complete sample from 1990Q1 through 2024Q1.

4.5 Dynamic Mode Decomposition

To estimate the autoregressive matrix B for the high-dimensional observable state, we use dynamic mode decomposition (DMD) to obtain a low-rank representation of its quarterly dynamics (Schmid, 2010, Kutz et al., 2016, Sargent et al., 2026b).

Three considerations motivate dimension reduction. First, under private-income ranking, there are fewer time observations than the $1 + 3M = 301$ components of the observable state, so an unrestricted autoregressive matrix B is not identified. The $1 + M = 101$ dimensional consumption-ranking system is also large relative to its quarterly sample. Second, the James–Stein phenomenon suggests that in high dimensions a biased, shrunk estimator can have lower mean-squared error than an unrestricted estimator. Third, the cost of solving the nonlinear robust-control problem grows rapidly with the state dimension, making reduction necessary for computational feasibility.

DMD estimates a rank-restricted B in the full observable state. Factoring this estimator gives a sufficient predictor for the next-state distribution. The fitted model’s nonlinear continuation problem then reduces exactly to this predictor, retaining the full residual covariance.

Writing n_y for the dimension of the observable state, the exact rank- N DMD estimator $\hat{B}$ is an $n_y \times n_y$ matrix of rank at most N . Appendix G.1 gives the truncated-singular-value construction. We estimate $\hat{\Omega} = \hat{\Gamma}\hat{\Gamma}^\top$ from the full-state residuals without truncating its rank. The sample size bounds its rank. For Cases 0 and 1, this step is all we need. Their value functions are affine, so we compute them on the fitted full-state model using linear algebra, at a cost polynomial in n_y .

For Cases 2 and 3, a grid over the full observable state is computationally infeasible. Let U_N contain the N retained left singular vectors and write the estimator as $\hat{B} = KU_N^\top$, where $K := Z_+ V_N \Sigma_N^{-1}$ uses the sample matrices in Appendix G.1. Define

$$x_t := U_N^\top y_t, \quad \tilde{B} := U_N^\top K = U_N^\top \hat{B} U_N, \quad Q_x := U_N^\top \hat{\Omega} U_N = LL^\top.$$

Then $\hat{B}y_t = Kx_t$, so the full fitted conditional law is $y_{t+1} = Kx_t + \hat{\Gamma}\eta_{t+1}$. In particular, it depends on y_t only through $x_t \in \mathbb{R}^N$. Its reduced marginal process is

$$x_{t+1} = \tilde{B}x_t + L\xi_{t+1}, \quad \xi_{t+1} \sim N(0, I_r), \quad r := \text{rank}(Q_x), \quad (59)$$

where $L \in \mathbb{R}^{N \times r}$ is a full-column-rank covariance factor. The eigenvalues of $\tilde{B}$ determine persistence, and L maps the r independent retained shocks into the N coordinates. The k -dimensional model shock of (1) is ε_{t+1} , and the shock to the fitted full state in (54) is

η_{t+1} . Appendix G.1 gives the sample matrices used to construct U_N , $\widehat{B}$, $\widehat{\Omega}$, $\widehat{\Gamma}$, and L . Current consumption remains $c_{q,t} = Y_t + \nu_q^{(c)} + e_q^{(c)} y_t$. Discarded current coordinates enter utility affinely, and all predictive dependence runs through x_t . We use the full conditional mean Kx_t in the continuation problem. Appendix G.2 proves the exact value representation and integrates the remaining Gaussian residual analytically.

We choose the common truncation rank $N = 5$ by expanding-window forecast accuracy. Figure 7 in Appendix I reports the retained variance share and the one-step-ahead RMSE for the consumption coordinates of y_{t+1} across alternative ranks. Five components minimize this forecast error under both rankings and retain about 95% of the variation under consumption ranking and 60% under private-income ranking.

All four cases use the same fitted full-state law and residual covariance $\widehat{\Omega}$. Only the nonlinear part of the value function requires a grid. For destination j , define the column vector d_j and the conditional Gaussian coefficients

$$d_j^\top := e_1^\top + (1 - \beta)e_j^{(c)}, \quad q_j := U_N^\top \widehat{\Omega} d_j, \quad (60)$$

$$a_j := L^+ q_j, \quad \sigma_{\perp,j}^2 := d_j^\top \widehat{\Omega} d_j - \|a_j\|^2 \geq 0, \quad (61)$$

where L^+ is the Moore–Penrose inverse. The range condition $q_j \in \text{Range}(L)$ holds because these quantities come from one joint covariance matrix. Conditional on destination j , the payoff-relevant innovation has the representation

$$d_j^\top \widehat{\Gamma} \eta = a_j^\top \xi + \sigma_{\perp,j} z, \quad z \sim N(0, 1), \quad z \perp \xi.$$

We evaluate the nonlinear expectation over ξ with the correlated term $a_j^\top \xi$ included and integrate the independent residual z analytically. The full Gaussian continuation law therefore requires at most N numerical integration dimensions. In the affine cases the resulting variance is exactly $\mathbf{e}_j = \|s_j\|^2$ from Proposition 5.

The variance share retained from Z_- describes variation in the observed state. The conditional variance $\sigma_{\perp,j}^2$ measures the payoff-relevant innovation remaining after conditioning on the retained shock ξ . Appendix H describes the grid and quadrature used to evaluate these Gaussian expectations.

Stability of the estimated dynamics can be checked directly. The exact DMD estimator $\widehat{B}$ has rank at most N , and its nonzero eigenvalues are the nonzero eigenvalues of $\widetilde{B}$, so the two matrices share a spectral radius. Under consumption ranking, $\rho(\widehat{B}) = \rho(\widetilde{B}) = 0.900$, and the next-largest eigenvalue moduli are 0.543 (a complex pair) and 0.413. Under private-income ranking, the spectral radius is 0.981, and the next-largest moduli are 0.919 and 0.746. Both spectral radii lie inside the unit circle, so the estimated A is stable as defined in

Section 2.1.2. The discounted welfare series of Appendix D are finite, and $\rho(G) = 1 < 1/\beta$ in Proposition 13.

4.6 Calibrating robustness via detection-error probabilities

We calibrate the penalty parameters θ_ε and θ_Π in (15) using a detection-error probability, which measures how reliably an observer could distinguish the approximating model from the worst-case model in a sample of fixed length (Anderson et al., 2003, Barillas et al., 2009). A common target equalizes statistical detectability across the two distortions. Case 1 calibrates θ_ε , Case 2 calibrates θ_Π , and Case 3 imposes $\theta_\varepsilon = \theta_\Pi = \theta_{\text{joint}}$ and calibrates the common value.

We compare the approximating kernel Q of (5) with the worst-case kernel $\tilde{Q}$ that the Bellman equation generates at a candidate penalty. Each simulated period records three components: a percentile transition $p_t \rightarrow p_{t+1}$, an innovation ξ_{t+1} to the reduced state, and the standardized conditional residual z_{t+1} for the realized destination from Section 4.5. These outcomes determine the fitted model's likelihood ratio. The sample log-likelihood ratio $\mathcal{L}_T$ of $\tilde{Q}$ relative to Q is

$$\mathcal{L}_T = \sum_{t=0}^{T-1} \Delta \mathcal{L}_{t+1}, \quad \Delta \mathcal{L}_{t+1} = \underbrace{\log \frac{\tilde{\Pi}_{p_t p_{t+1}}(x_t)}{\Pi_{p_t p_{t+1}}}}_{\text{transition}} + \underbrace{\log \frac{\tilde{\varphi}_{p_{t+1}}(\xi_{t+1} | x_t)}{\varphi(\xi_{t+1})}}_{\text{innovation}} + \underbrace{\log g_{\perp, p_{t+1}}(z_{t+1})}_{\text{conditional Gaussian residual}}. \quad (62)$$

Case 1 holds $\tilde{\Pi} = \Pi$ and drops the transition score, Case 2 sets $\theta_\varepsilon = \infty$ and drops the other two, and Case 3 keeps all three.

The residual z is standard normal under Q and has mean $-\mathbf{m}_{\perp, j}$ under $\tilde{Q}$, where $\mathbf{m}_{\perp, j} := \beta \sigma_{\perp, j} / \theta_\varepsilon$ and $\log g_{\perp, j}(z) = -\mathbf{m}_{\perp, j} z - \mathbf{m}_{\perp, j}^2 / 2$. We tilt the retained innovation density by the full payoff $\hat{v}_j(\tilde{B}x + L\xi) + a_j^\top \xi$, including its correlated linear term. In Cases 2 and 3 the transition and innovation likelihoods therefore come from the Bellman equation. Appendix H.2 gives the likelihood ratios used in the quadrature approximation. Appendix H.3 also gives an exact Gaussian Case 1 formula as a check on the detection-error calculation.

We define the detection-error probability as

$$\bar{p}(\theta) := \frac{1}{2} [\text{Prob}(\mathcal{L}_T > 0 | Q) + \text{Prob}(\mathcal{L}_T \leq 0 | \tilde{Q})], \quad (63)$$

the probability that a likelihood-ratio classifier with equal priors picks the wrong model. A value near one half describes two models that a record of length T cannot separate.

We set $T = 137$, the number of quarters we observe. Both simulations start from a

common draw of p_0 from the stationary distribution of Π and x_0 from the stationary distribution of the reduced state. Setting T to the length of our sample matters because $\bar{p}$ falls with T . Holding $\bar{p}$ fixed shrinks the calibrated distortion as the historical record lengthens, at rate $T^{-1/2}$ in the equal-variance Case 1 Gaussian benchmark. Appendix H.3 derives that rate from the worst-case mean shift of Proposition 5.

The comparison between the two doubts turns on whose information defines detectability. The classifier in (63) sees one simulated history containing T realizations of the innovation and T realizations of a single household's percentile transitions. A household likewise observes T draws of the common innovation and T percentile transitions along its own path (Section 2.3).

We estimate Π from many thousands of linked interviews, so the pooled sample can detect distortions that are hard to distinguish from the approximating model along a single household's 137-quarter history. A calibration based on the sampling distribution of $\hat{\Pi}$ would allow smaller transition distortions and imply a smaller transition contribution to welfare costs. We calibrate both components using the household's own history.

For each case we bisect on $\log \theta$ until $\bar{p}(\theta) = \bar{p}_{\text{target}}$. At each candidate penalty, we solve the corresponding Bellman equation, form the worst-case kernel, and simulate 2,000 paths under each model. We use common random numbers across candidates to reduce simulation noise and stabilize the bisection. Near $\bar{p} = 0.45$, 2,000 paths give a Monte Carlo standard error of about 0.008 for the averaged estimate. In every case, we find that $\bar{p}$ increases in θ over the search bracket. Algorithm 1 and Appendices H.1 and H.3 give the search and simulation settings.

Our baseline target is $\bar{p}_{\text{target}} = 0.45$, a value at which the classifier identifies the generating model only 55% of the time.

4.7 Welfare calculations

We compute the four value functions and convert them into compensating differences. Proposition 5 gives Cases 0 and 1 in closed form. For Cases 2 and 3, Proposition 13 establishes existence, uniqueness, and convergence of value-function iteration for the Bellman operator on the unbounded state space. We implement these cases on a finite grid for the reduced state x . The Bellman operator implemented on that grid, with the quadrature and interpolation rules of Appendix H.1, is a β -contraction and therefore has a unique fixed point. We set the quarterly discount factor to $\beta = 0.99$. That appendix also describes the grid, the integration rule, and the resulting limits on numerical accuracy.

Every welfare number compares two values at the same state: the risky value $v_p^{(m)}$ of (18)

under Case m and the benchmark w_p of (38), which removes innovations in percentile-mean consumption while preserving mobility. Their gap is (41). On the fitted model, current consumption is $c_p = Y + \nu_p^{(c)} + e_p^{(c)}y$ and the next location index satisfies $Y' = Y + \nu + e_1^\top y'$. The exact value representation is

$$v_p^{(m)}(Y, y) = Y + (1 - \beta)e_p^{(c)}y + \frac{\beta\nu}{1 - \beta} + \hat{v}_p^{(m)}(U_N^\top y). \quad (64)$$

Translation equivariance accounts for the term Y . If $U_N^\top h = 0$, the states y and $y + h$ have the same next-state distribution, so their values differ only by the current utility term $(1 - \beta)e_p^{(c)}h$. The remaining nonlinear dependence therefore factors through $x = U_N^\top y$. The constant $\beta\nu/(1 - \beta)$ removes the discounted deterministic drift. Appendix G.2 gives the proof and shows that this representation preserves the unique linear-growth fixed point.

With d_j , a_j , and $\sigma_{\perp,j}$ defined in Section 4.5, the reduced value solves

$$\hat{v}_p(x) = (1 - \beta)\nu_p^{(c)} - \theta_\Pi \log \left[\sum_{j=0}^{M-1} \Pi_{pj} \exp \left(-\frac{(\mathcal{R}_\varepsilon \hat{v}_j)(x)}{\theta_\Pi} \right) \right], \quad (65)$$

$$\begin{aligned} (\mathcal{R}_\varepsilon \hat{v}_j)(x) &= \beta d_j^\top Kx - \frac{\beta^2 \sigma_{\perp,j}^2}{2\theta_\varepsilon} \\ &\quad - \theta_\varepsilon \log \mathbb{E}_\xi \exp \left[-\frac{\beta}{\theta_\varepsilon} \left\{ \hat{v}_j(\tilde{B}x + L\xi) + a_j^\top \xi \right\} \right]. \end{aligned} \quad (66)$$

where we suppress the case index. Each case fixes the pair $(\theta_\varepsilon, \theta_\Pi)$, and a logarithmic expression is replaced by the expectation it converges to when its penalty is infinite. In particular, for Case 2, $(\mathcal{R}_\varepsilon \hat{v}_j)(x) = \beta d_j^\top Kx + \beta \mathbb{E}_\xi \hat{v}_j(\tilde{B}x + L\xi)$. For finite θ_ε , the term $a_j^\top \xi$ preserves the covariance between the payoff-relevant innovation and the next reduced state. The variance term integrates only the conditionally independent residual. Together these terms retain the future common innovation as well as the destination-specific consumption innovation.

We compute the compensating differences directly from the four value functions. The explicit terms in (64) are the same in every case, so $v_p^{(m)} - v_p^{(0)} = \hat{v}_p^{(m)} - \hat{v}_p^{(0)}$. Since $w_p - v_p^{(0)} = \alpha_p^{(0)}$ from (42),

$$\alpha_p^{(m)}(x) = \alpha_p^{(0)} - [\hat{v}_p^{(m)}(x) - \hat{v}_p^{(0)}(x)]. \quad (67)$$

Cases 0 and 1 deliver one number per percentile, whereas Cases 2 and 3 deliver a function of the state. To compare them, we average over the stationary Gaussian distribution ϖ of

the approximating reduced-state process (59) and report

$$\bar{\alpha}_p^{(m)} := \int \alpha_p^{(m)}(x) \varpi(\mathrm{d}x) = \mathbb{E}_\varpi[\alpha_p^{(m)}(x_t)] , \quad m = 2, 3, \quad (68)$$

approximating the integral with the stationary Gaussian density at the grid points when its covariance is nonsingular, and integrating on its Gaussian support otherwise. For both cases, we average over x_t using this distribution at each fixed current percentile p .

The compensating difference is an additive shift to every component of c_t at every date, and we report it in two units. The *level equivalent*

$$\text{LE}_p^{(m)} := 100 \bar{\alpha}_p^{(m)} \quad (69)$$

states the shift in log percentage points, which for small shifts is approximately the percentage increase in consumption. This is the unit of the 0.05% consumption equivalent in Lucas (1987). The *growth-rate equivalent* instead expresses the same shift as a permanent increase in annual log-consumption growth of equal present value. A quarterly growth increment δ has normalized present value $\beta\delta/(1-\beta)$, so equating that to $\bar{\alpha}_p^{(m)}$ and annualizing gives

$$\text{GE}_p^{(m)} := \frac{400(1-\beta)}{\beta} \bar{\alpha}_p^{(m)}, \quad (70)$$

measured in annual percentage points. At $\beta = 0.99$ the conversion factor is $400(1-\beta)/\beta \approx 4.04$, so a level equivalent is about 24.75 times the corresponding growth-rate equivalent. The tables report GE, and we quote selected LE values in the text.

5 Results

We first measure the cost of consumption risk when households trust the estimated model, then add doubts about innovations, mobility, and both. We compare the resulting welfare costs across percentiles under consumption and private-income ranking and examine how they change with the detection-error target.

Throughout, welfare costs are growth-rate equivalents $\text{GE}_p^{(m)}$ (70) for case $m \in \{0, 1, 2, 3\}$, in annual percentage points. For each of Cases 1–3, we recalibrate the relative entropy penalty parameters to attain the detection-error target $\bar{p} = 0.45$. The cases differ in which part of the model the household treats as potentially misspecified.

5.1 Case 0: the cost of consumption risk

Case 0 is a benchmark in which the household trusts both the estimated distribution of the innovations and the estimated transition matrix Π . Its compensating difference measures the welfare cost of risk in the estimated percentile-mean consumption process.

We calculate this cost in two ways. In our first calculation, households move across percentiles under the estimated transition matrix. In the second, we set $\Pi = I$, so that a household remains at its current percentile. We hold the matrices (A, C, D, F) fixed across the two calculations.

Table 1 reports both calculations as annual growth-rate equivalents. Under estimated mobility, the cost is about one tenth of an annual percentage point of consumption growth. At the reported percentiles, it ranges from 0.0855 to 0.0857 under consumption ranking and from 0.1024 to 0.1048 under private-income ranking.

Under consumption ranking, estimated mobility raises the cost by 0.0015 annual percentage points at the 10th percentile and by 0.0022 at the median. It changes the cost by less than 0.0001 at the 90th percentile. Under private-income ranking, the fixed-percentile cost falls from 0.1117 at the 10th percentile to 0.0909 at the 90th percentile. Estimated mobility lowers the cost at the 10th percentile, raises it at the median and the 90th percentile, and substantially reduces its variation across percentiles.

Table 1: Case 0 welfare costs with and without estimated mobility. Entries are growth-rate equivalents $\text{GE}_p^{(0)}$ in annual percentage points. The final column is the estimated- Π value minus the fixed-percentile value. Differences are calculated before rounding.

Percentile	Fixed percentiles ($\Pi = I$)	Estimated Π	Difference
<i>Panel A: consumption ranking</i>			
10th	0.0843	0.0857	+0.0015
50th	0.0833	0.0855	+0.0022
90th	0.0856	0.0855	−0.0001
<i>Panel B: private-income ranking</i>			
10th	0.1117	0.1048	−0.0069
50th	0.0981	0.1034	+0.0053
90th	0.0909	0.1024	+0.0115

Equation (42) explains why mobility changes the cost at a given percentile. Let $\sigma_h^2 = (\sigma_{0,h}^2, \dots, \sigma_{M-1,h}^2)^\top$ collect the conditional variances of log consumption at horizon h . For a

household that starts at percentile p ,

$$[\Pi^h \sigma_h^2]_p = \sum_{j=0}^{M-1} (\Pi^h)_{pj} \sigma_{j,h}^2,$$

where $(\Pi^h)_{pj}$ is the probability that the household is at percentile j after h periods.

The difference between the two Case 0 costs is therefore

$$\alpha_p^{(0)}(\Pi) - \alpha_p^{(0)}(I) = \frac{1-\beta}{2} \sum_{h=1}^{\infty} \beta^h \left\{ \sum_{j=0}^{M-1} (\Pi^h)_{pj} \sigma_{j,h}^2 - \sigma_{p,h}^2 \right\}.$$

The larger changes under private-income ranking reflect the greater variation in horizon-specific conditional variances across percentiles together with the transition probabilities under that ranking.

Mobility changes the cost at each percentile but leaves one weighted mean unchanged. Let π be the stationary distribution of the estimated transition matrix. Corollary 7 gives

$$\sum_{p=0}^{M-1} \pi_p \alpha_p^{(0)}(\Pi) = \sum_{p=0}^{M-1} \pi_p \alpha_p^{(0)}(I).$$

The equality holds across all 100 percentiles with (A, C, D, F) and the variance vectors σ_h^2 fixed. Each weight π_p is the long-run fraction of time a household moving under the estimated Π spends at percentile p . The equality compares its long-run average cost with the fixed-percentile costs averaged using the same weights.

Figure 3 shows the fixed-percentile and estimated-mobility calculations for all 100 percentiles. It also separates each cost into an i.i.d.-risk component and a persistent-risk component. Following Sargent et al. (2026a, Section 5.2), we decompose each horizon-specific variance into the contribution of the most recent innovation and the contributions of earlier innovations. For each percentile j and horizon h ,

$$\sigma_{j,h}^2 = \|F_j\|^2 + \sum_{i=1}^{h-1} \|J_{j,i}\|^2 = \|F_j\|^2 + (\sigma_{j,h}^2 - \|F_j\|^2).$$

The first term is the variance from the innovation that arrives in the final quarter of the horizon. Summing this term over horizons defines the i.i.d.-risk component. If

$$\mathbf{f} := (\|F_0\|^2, \dots, \|F_{M-1}\|^2)^\top,$$

then the two components are

$$\begin{aligned}\alpha_{p,\text{iid}}^{(0)} &:= \frac{(1-\beta)\beta}{2} [\Pi(I - \beta\Pi)^{-1}\mathbf{f}]_p, \\ \alpha_{p,\text{persistent}}^{(0)} &:= \alpha_p^{(0)} - \alpha_{p,\text{iid}}^{(0)}.\end{aligned}$$

When $\Pi = I$, the i.i.d.-risk component reduces to $\frac{\beta}{2}\|F_p\|^2$. Every innovation ε_t is independent and identically distributed over time. An earlier innovation has loading

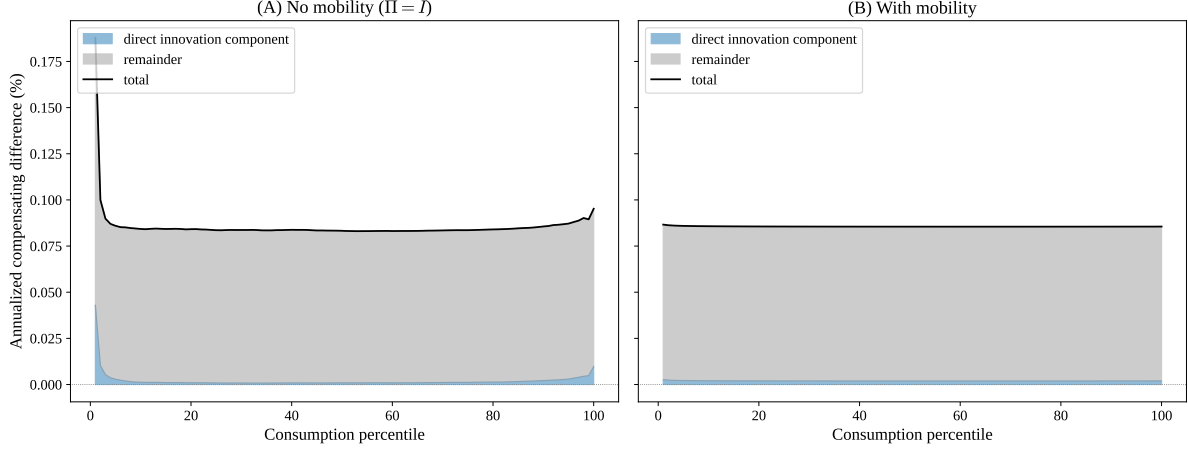
$$J_{j,i} = F_j + \sum_{r=0}^{i-1} D_j A^r C, \quad i \geq 1.$$

The loading $J_{j,i}$ combines an innovation's lasting effect on the consumption level with its propagation through the state. The persistent-risk component can therefore be positive even when $D_j = 0$.

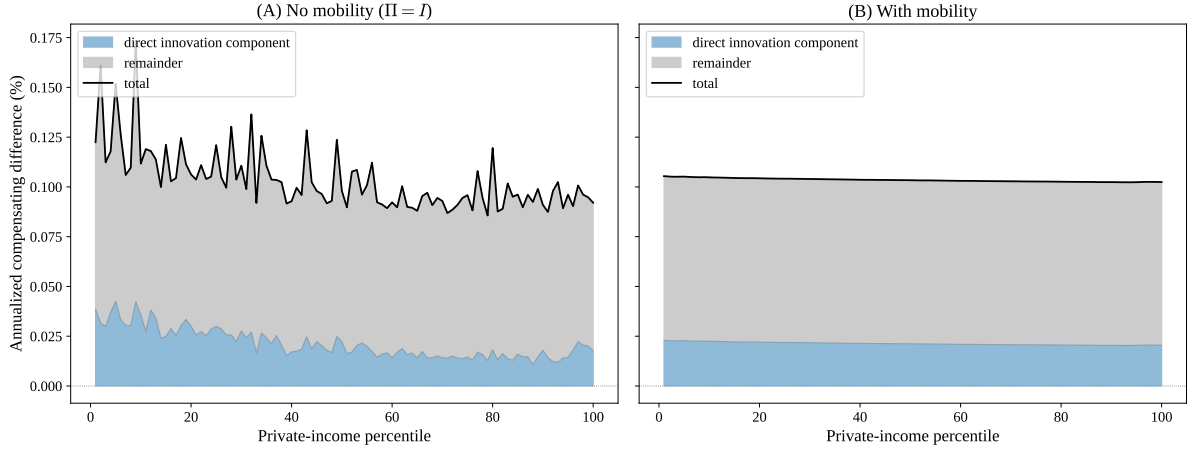
The left side of each panel in Figure 3 holds a household's percentile fixed. Under consumption ranking, the cost is nearly level through the middle of the distribution and rises at both ends. The increase is especially large at the bottom, where the cost at the 1st percentile is more than twice the cost near the middle. Under private-income ranking, the fixed-percentile cost generally declines from the bottom to the top and also changes sharply between some neighboring percentiles.

The right side of each panel allows mobility under the estimated Π . The powers Π^h average the horizon-specific variances over the percentiles a household may reach. This averaging reduces the large differences at the ends of the distribution, the changes between neighboring percentiles, and most of the downward slope under private-income ranking.

Persistent risk accounts for most of the Case 0 cost in the estimated model. At the 10th, 50th, and 90th percentiles, it accounts for about 98 percent of the Case 0 cost under consumption ranking and about 79–80 percent under private-income ranking.



(a) Consumption ranking.



(b) Private-income ranking.

Figure 3: How mobility under the estimated transition matrix changes Case 0 welfare costs across the distribution. The blue area, labeled the direct innovation component in the panels, is the i.i.d.-risk component: the contribution of the innovation arriving in the final quarter of a forecast horizon. The gray remainder is the persistent-risk component, contributed by earlier innovations that still affect the consumption level.

5.2 Calibrated penalty parameters

Table 2 reports the penalties we calibrate to the detection-error target $\bar{p} = 0.45$ for Cases 1–3 (Section 4.6).

The common Case 3 penalty exceeds both separately calibrated penalties. Both distortions contribute to the likelihood ratio in Case 3, so a larger penalty is needed to keep the joint worst-case model as hard to detect as either model with a single source of uncertainty. At $x = 0$, transition distortions account for 57 percent of Case 3 relative entropy under consumption ranking and 55 percent under private-income ranking.

Table 2: Penalties calibrated to the detection-error target $\bar{p} = 0.45$. An entry of ∞ marks a component the household fully trusts. Case 3 imposes a common penalty $\theta_\varepsilon = \theta_\Pi$ on both components.

	Consumption ranking		Private-income ranking	
	θ_ε	θ_Π	θ_ε	θ_Π
Case 1: innovation uncertainty	1.0288	∞	0.8466	∞
Case 2: transition uncertainty	∞	0.9268	∞	0.7953
Case 3: joint uncertainty	1.4423	1.4423	1.3875	1.3875

5.3 The worst-case transition matrix

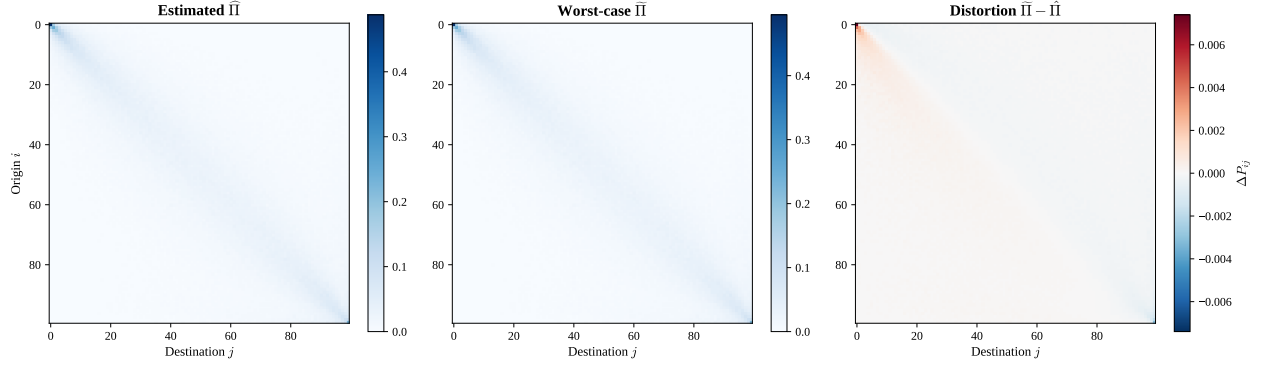
At the calibrated penalty parameters of Table 2, Figure 4 shows the Case 3 worst-case transition matrix, evaluated at the stationary mean $x = 0$ of the reduced state, beside the estimated matrix and the difference between the two.

Under both rankings, the worst-case matrix assigns more probability to destinations with lower continuation values. Staying probabilities rise at low percentiles and fall at high percentiles, shifting the stationary distribution toward the bottom. The estimated private-income matrix puts more probability on the diagonal than the consumption matrix does, but both allow moves across a wide range of percentiles.

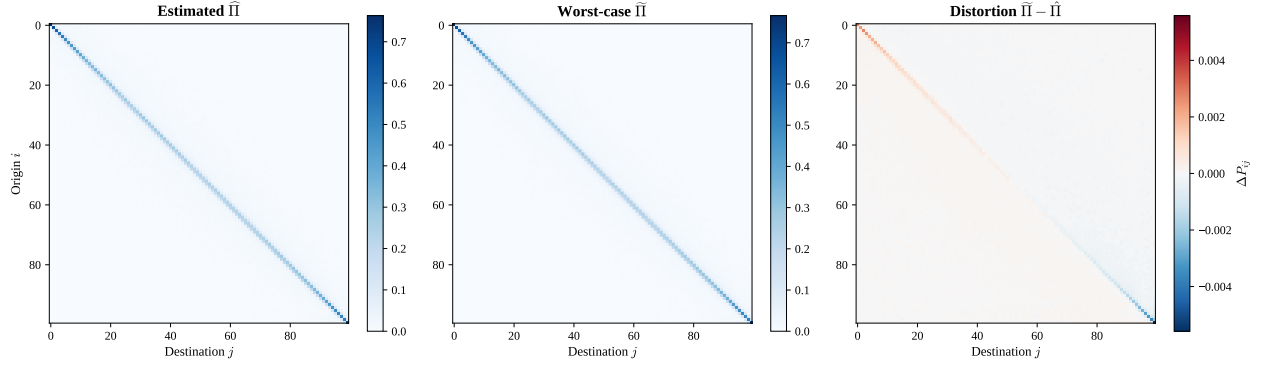
The probability changes in Figure 4 are small in a single quarter, but they affect the continuation value at every future date. The second-largest eigenvalue modulus of the estimated transition matrix is 0.85 under consumption ranking and 0.93 under private-income ranking. Deviations from the stationary distribution along the corresponding modes have approximate half-lives of four and nine quarters. In addition, the household discounts the future at $\beta = 0.99$, and the robust Bellman equation applies the conditional transition distortion in every period.

Figure 5 compares the stationary distributions and staying probabilities of the estimated and worst-case matrices. We evaluate the Case 3 worst-case matrix at $x = 0$ and then hold it fixed. The stationary distributions show how the changes in transition probabilities accumulate over time.⁷ Under both rankings, the worst-case stationary distribution places more weight on low percentiles and less on high percentiles. The shift is larger under private-income ranking.

⁷The estimated stationary distribution is close to uniform (Section 4.2). Cross-sectional percentile shares are $1/M$ by construction.

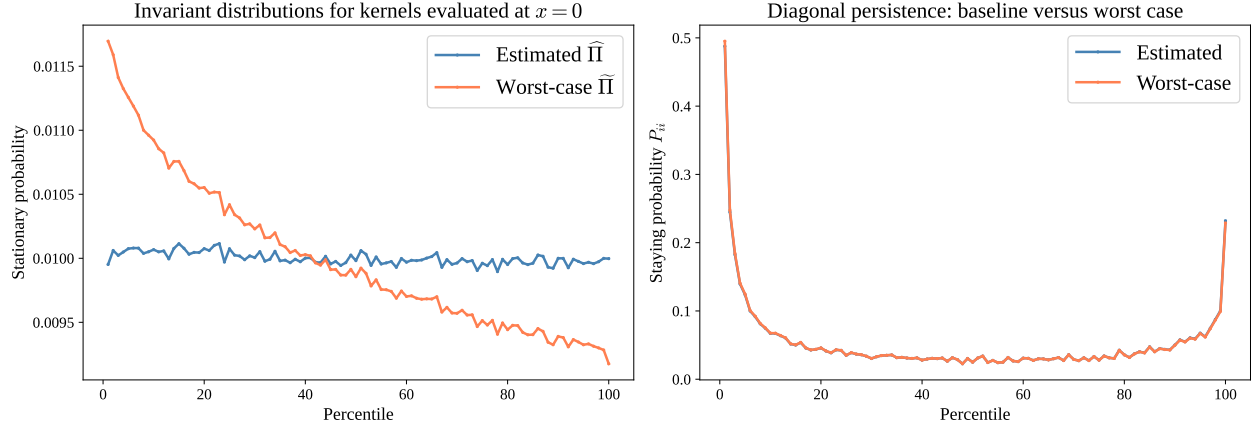


(a) Consumption ranking.

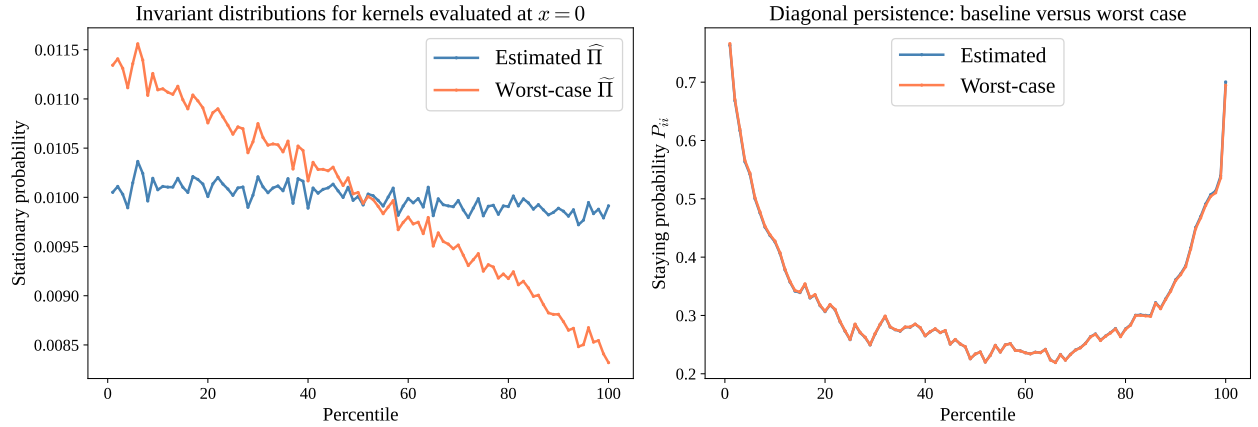


(b) Private-income ranking.

Figure 4: Worst-case transition matrices in Case 3 at $\bar{p} = 0.45$, evaluated at the stationary mean $x = 0$ of the reduced state. Within each panel: the estimated matrix (left), the worst-case matrix (center), and their difference (right), drawn on a finer color scale.



(a) Consumption ranking.



(b) Private-income ranking.

Figure 5: Stationary distributions and one-period stay probabilities for Case 3 at $\bar{p} = 0.45$. The left side of each panel shows the stationary distributions of the estimated transition matrix and the worst-case matrix evaluated and held fixed at $x = 0$. The right side shows the corresponding probabilities of remaining at the current percentile. Each stationary distribution describes the long-run fraction of time that one household spends at each percentile under the corresponding fixed matrix.

5.4 Welfare costs of consumption risk and model uncertainty

Table 3 compares welfare costs across the four cases at the calibrated penalties in Table 2. For each current percentile p , the first four columns report the permanent increase in annual log-consumption growth that has the same present value as the compensating change in the consumption level.

The compensating differences in Cases 2 and 3 vary with the reduced state. For these cases, the table reports the stationary reduced-state average defined in (68). The final three columns report $\Delta_{p,\varepsilon}$, $\Delta_{p,\Pi}$, and their ratio, where

$$\Delta_{p,\varepsilon} := \text{GE}_p^{(1)} - \text{GE}_p^{(0)}, \quad \Delta_{p,\Pi} := \text{GE}_p^{(2)} - \text{GE}_p^{(0)}.$$

Table 3: Welfare costs and additional costs from model uncertainty at $\bar{p} = 0.45$. Entries are growth-rate equivalents $\text{GE}_p^{(m)}$ in annual percentage points. Cases 0 and 1 use the closed forms of Propositions 6 and 5. Cases 2 and 3 come from value-function iteration and are averaged over the stationary distribution of the reduced state.

Percentile p	Welfare cost				Additional cost relative to Case 0		
	Case 0	Case 1	Case 2	Case 3	$\Delta_{p,\varepsilon}$	$\Delta_{p,\Pi}$	$\Delta_{p,\Pi}/\Delta_{p,\varepsilon}$
<i>Panel A: consumption ranking</i>							
10th	0.0857	0.1654	0.2097	0.2207	0.0797	0.1240	1.56
50th	0.0855	0.1652	0.2052	0.2178	0.0797	0.1197	1.50
90th	0.0855	0.1653	0.2039	0.2169	0.0797	0.1183	1.48
<i>Panel B: private-income ranking</i>							
10th	0.1048	0.1868	0.2079	0.2139	0.0820	0.1031	1.26
50th	0.1034	0.1853	0.2089	0.2137	0.0819	0.1056	1.29
90th	0.1024	0.1843	0.2114	0.2146	0.0819	0.1090	1.33

Joint uncertainty produces the largest welfare cost: about 0.22 annual percentage points under consumption ranking and 0.21 under private-income ranking, compared with Case 0 costs of about 0.086 and 0.10. At the median, joint uncertainty raises the cost to 2.5 and 2.1 times its Case 0 value. Welfare costs rise from Case 0 through Case 3 at all 100 percentiles under both rankings.

Transition uncertainty has the larger of the two separate effects. At the 10th consumption percentile, the innovation increment is 0.0797 annual percentage points and the transition increment is 0.1240, a ratio of 1.56. The ratios at the median and 90th percentile are 1.50 and 1.48. Under private-income ranking, the corresponding ratios are 1.26, 1.29, and 1.33.

Multiplying the growth-rate equivalents by 24.75 gives the costs in log percentage points

of consumption (Section 4.7). At the median, the Case 0 level-equivalent cost is $LE_{50}^{(0)} = 2.12$ log percentage points under consumption ranking and 2.56 under private-income ranking. Joint uncertainty raises these costs to 5.39 and 5.29 log percentage points, respectively.

These level equivalents are comparable to the 0.05 percent figure in Lucas (1987). Our Case 0 costs are about forty to fifty times larger. Lucas measures the welfare gain from removing transitory deviations of aggregate consumption from a deterministic trend. Our benchmark w_p in (38) replaces consumption at every horizon and future percentile by its conditional mean in levels. Thus $\alpha^{(0)}$ measures the welfare gain from removing innovations in percentile-mean consumption, including the martingale component of Section 4.3. That component makes $\sigma_{p,h}^2$ grow linearly with the horizon and increases the discounted sum in (42). This is the mechanism of Obstfeld (1994) and Tallarini (2000). Consistent with the importance of persistent consumption risk, the persistent-risk component in Section 5.1 accounts for about 98 percent of the Case 0 cost under consumption ranking.

The larger common penalty in Case 3 restrains its transition distortion relative to Case 2. The joint cost still exceeds Case 2 by 0.0110–0.0131 annual percentage points under consumption ranking and 0.0032–0.0059 under private-income ranking. Appendix J compares this calibration with the joint problem at the separate Case 1 and Case 2 penalties.

5.5 Why the cost varies across percentiles

Under consumption ranking, the Case 2 cost falls from 0.2097 at the 10th percentile to 0.2039 at the 90th, and the Case 3 cost falls from 0.2207 to 0.2169. Under private-income ranking, Case 2 rises from 0.2079 to 0.2114, while Case 3 stays close to 0.214. The Case 0 and Case 1 costs vary little across the reported percentiles under either ranking.

Equation (19) tilts each row of the estimated transition matrix toward destinations with lower continuation values. The welfare loss depends on the probabilities in that row and the differences in continuation values across the destinations it can reach.

Under consumption ranking, the estimated matrix spreads probability across a wide range of destinations. Differences in expected continuation values across those destinations are larger near the bottom, where the transition-uncertainty increment is also largest among the reported percentiles. Under private-income ranking, the matrix concentrates probability on and near the diagonal. The relevant differences in continuation values are larger at the 90th percentile than at the 10th, as is the transition increment. Section 5.6 measures these exposures directly.

5.6 A first-order account of the two doubts

The first-order expansion in Proposition 3 explains the relative sizes of the two uncertainty costs and their variation across percentiles using quantities from the estimated model. It separates the variance of discounted continuation value $Y_j = \beta v_j(\zeta', c')$ into two components, where j is the household's next-period percentile.

The first component is $D_p = \mathbb{E}_{\Pi_p}[\text{Var}_\varepsilon(Y_j)]$, within-destination variation associated with uncertainty about the innovation distribution. The second is $M_p = \text{Var}_{\Pi_p}(\mathbb{E}_\varepsilon[Y_j])$, between-destination variation associated with uncertainty about the transition probabilities.

These one-period adjustments also affect later continuation values. Let $S := (I - \beta\Pi)^{-1}$ and let ϖ be the stationary distribution of the approximating reduced state. At the Case 0 value function, D is state independent, while M generally depends on x . Write $\overline{M} := \int M(x) \varpi(dx)$. Stationarity and independence of the approximating state dynamics from the percentile chain imply $\int \sum_j \Pi_{pj} \mathbb{E}[f_j(x') \mid x] \varpi(dx) = \sum_j \Pi_{pj} \int f_j(x) \varpi(dx)$. Thus S propagates the stationary-averaged first-order adjustment, giving

$$\alpha^{(1)} - \alpha^{(0)} = \frac{SD}{2\theta_\varepsilon}, \quad (71)$$

$$\overline{\alpha^{(2)} - \alpha^{(0)}} \approx \frac{S\overline{M}}{2\theta_\Pi}. \quad (72)$$

For percentile p , dividing the second expression by the first gives

$$\frac{\overline{\alpha_p^{(2)} - \alpha_p^{(0)}}}{\alpha_p^{(1)} - \alpha_p^{(0)}} \approx \frac{[S\overline{M}]_p}{[SD]_p} \frac{\theta_\varepsilon}{\theta_\Pi}. \quad (73)$$

A smaller penalty permits a larger distortion, so the relative penalties matter even when the two cases have the same detection-error target.

For the affine Case 1 value function with $\theta_\Pi = \infty$, the entropic expansion terminates at the variance term (Remark 4). Hence $SD/(2\theta_\varepsilon)$ equals the closed-form increment $-\kappa$ of (36). The Case 2 approximation omits higher conditional cumulants from the outer certainty equivalent.

Table 4 reports the two variance terms and the welfare effects they imply, using the same stationary distribution as the main welfare calculations.

Table 4: Stationary-averaged first-order approximation at the Case 0 value function. D_p is the within-destination conditional variance of discounted continuation value, and $\bar{M}_p$ is the stationary average of its between-destination conditional variance. The value function is measured in permanent log-consumption units, so multiplying the variances by 10^4 reports them in squared log percentage points. The last two columns report the first-order effects $[SD]_p/(2\theta_\varepsilon)$ and $[SM]_p/(2\theta_\Pi)$, converted by (70) to annual percentage points.

Percentile	Variance ($\times 10^4$)		First-order welfare effect	
	D_p	$\bar{M}_p$	Innovation	Transition
<i>Panel A: consumption ranking</i>				
10th	4.040	8.040	0.0797	0.1209
50th	4.061	4.500	0.0797	0.1166
90th	4.079	3.713	0.0797	0.1152
<i>Panel B: private-income ranking</i>				
10th	3.477	4.062	0.0820	0.1052
50th	3.416	4.215	0.0819	0.1074
90th	3.455	4.738	0.0819	0.1104

The last two columns use the separately calibrated Case 1 and Case 2 penalties. Their ratio $\theta_\varepsilon/\theta_\Pi$ is 1.11 under consumption ranking and 1.06 under private-income ranking. At the 10th percentile, the first-order transition-to-innovation ratio is 1.52 under consumption ranking and 1.28 under private-income ranking, compared with 1.56 and 1.26 in Table 3.

The innovation column reproduces the closed-form increments in Table 3. The first-order transition effect is about 0.0031 annual percentage points below the computed Case 2 increment at each reported consumption percentile. Under private-income ranking, it is 0.0014–0.0020 above the computed increment.

The between-destination variance $\bar{M}_p$ falls from the 10th percentile to the 90th under consumption ranking and rises under private-income ranking. The first-order transition effect follows the same pattern, though its variation is smaller because S averages the one-period exposures over future percentile transitions. The within-destination variance D_p and the innovation effect are nearly constant under both rankings. These exposures explain both the larger cost of transition uncertainty and the opposite slopes of its cost across the two rankings.

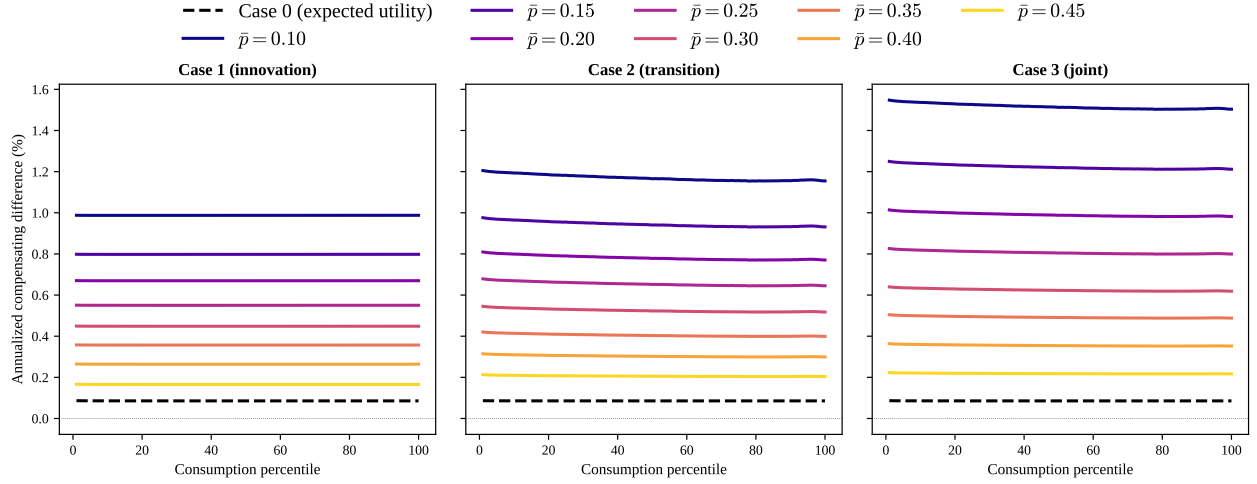
5.7 Sensitivity to the detection-error target

All results reported above use the detection-error target $\bar{p} = 0.45$. To assess how this choice affects the results, Figure 6 repeats the calibration for several values of $\bar{p}$ under both

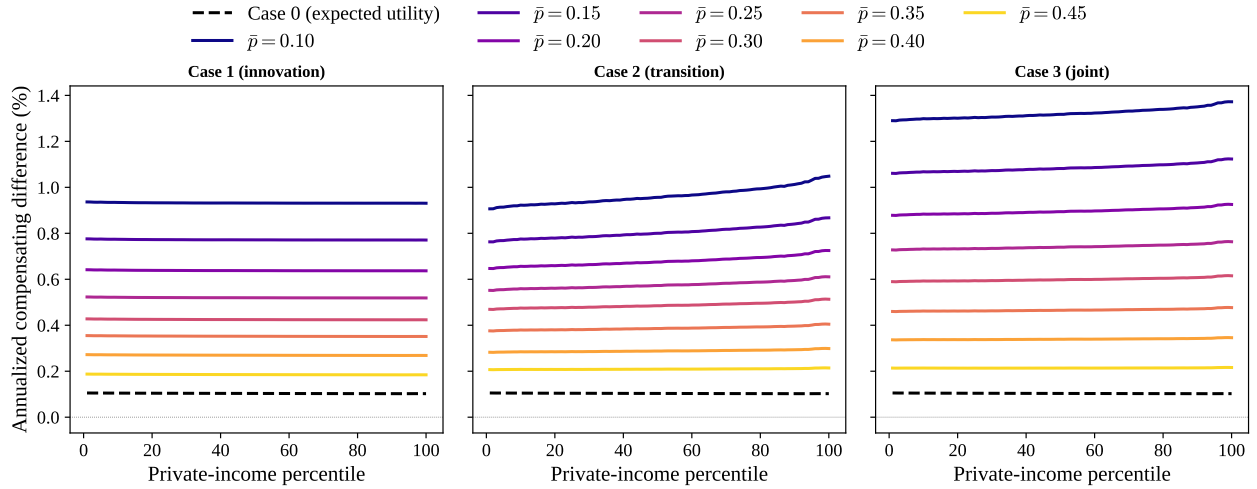
rankings. For each target, we recalibrate the relevant penalty parameter, or parameters, and then recompute the welfare cost.

A lower target requires the approximating and worst-case models to be easier to distinguish. In our calibration, this permits a larger departure from the approximating model and represents greater concern about model misspecification. As $\bar{p}$ falls, welfare costs rise in Cases 1–3. Case 0 retains its benchmark cost at every target. At $\bar{p} = 0.45$, the Case 3 cost is about 0.22 annual percentage points under consumption ranking and 0.21 under private-income ranking. At the lowest target shown, $\bar{p} = 0.10$, it reaches 1.50–1.54 at the reported consumption percentiles and 1.30–1.35 at the reported private-income percentiles.

The way costs vary with the current percentile also depends on the source of model uncertainty. The Case 1 curves are nearly horizontal at every target under both rankings. The Case 2 and Case 3 curves vary more across percentiles as the target falls. Under consumption ranking, costs are lower at the 90th percentile than at the 10th, with a small upturn near the top of the distribution. Under private-income ranking, costs are higher at the 90th percentile, and the upward slope becomes more pronounced as the target falls.



(a) Consumption ranking.



(b) Private-income ranking.

Figure 6: Welfare costs by percentile and detection-error target. For each ranking, the panels show Cases 1–3 from left to right. Each colored line corresponds to a detection-error target, with the relevant penalty parameters recalibrated at that target. The dashed line shows Case 0, which does not depend on the target. The plotted values are growth-rate equivalents $GE_p^{(m)}$ in annual percentage points. For Cases 2 and 3, we average the compensating difference over the stationary distribution of the reduced state. The curves for $\bar{p} = 0.45$ reproduce the values in Table 3.

5.8 Role of the ranking variable

The ranking variable changes both the estimated consumption dynamics and the transition matrix. Together, these changes produce the opposite patterns in the cost of transition uncertainty under the two rankings.

Households assigned to the same narrow percentile group have similar current consump-

tion by construction when the ranking variable is consumption, whereas under private-income ranking they can differ in taxes, transfers, saving, and consumption. These differences affect the horizon-specific conditional variances $\sigma_{p,h}^2$ and their variation across percentiles. At the 10th, 50th, and 90th percentiles, the fixed-percentile Case 0 costs are lower and more alike under consumption ranking (Table 1).

The transition matrix determines how these exposures follow a household as it moves across percentiles. As Figure 1 shows, consumption ranks are less persistent than private-income ranks. Both matrices admit moves across a wide range of destinations, and the worst-case model gives more weight to those with lower continuation values. The size of the welfare loss depends on the estimated probabilities and the differences in continuation values across those destinations.

6 Concluding Remarks

We study how model uncertainty changes the welfare cost of consumption risk for households that move across percentiles in income and consumption distributions. We distinguish doubts about the conditional distribution of consumption innovations from doubts about the transition probabilities that govern mobility across the distribution. Using the chain rule for relative entropy, we derive a nested multiplier recursion that assigns a separate penalty to each source. We then estimate both components from CEX data, defining percentiles once by consumption and once by private income.

In Case 0, estimated mobility averages consumption-risk exposures over the destinations a household may reach. With the estimated consumption dynamics and horizon-specific conditional variances held fixed, mobility changes costs at individual percentiles and preserves their π -weighted mean. Doubts about the transition matrix tilt probabilities toward reachable destinations with lower continuation values, breaking this neutrality.

Our Case 0 costs are close to those that Sargent et al. (2026a) report for households whose percentiles are fixed. We find that eliminating innovations in percentile-mean consumption while preserving mobility is worth about one tenth of an annual percentage point of growth, or 2.1–2.6 log percentage points of consumption at the reported percentiles. These costs are forty to fifty times Lucas’s 0.05 percent.

At $\bar{p} = 0.45$, joint doubts raise the cost from 0.086–0.105 to about 0.21–0.22 annual percentage points. The median level-equivalent cost rises to 5.39 log percentage points under consumption ranking and 5.29 under private-income ranking. The source of a doubt matters as much as its size. At a common detection-error target the transition increment is 1.3–1.6 times the innovation increment at the reported percentiles, and when both doubts

operate at that target the joint increment is 59–66 percent of the sum of the two separate ones.

Transition uncertainty accounts for most of the variation in the additional cost across current percentiles. From the 10th to the 90th percentile, the transition-uncertainty increment declines under consumption ranking and rises under private-income ranking, whereas the innovation-uncertainty increment is nearly flat under both rankings. Our first-order decomposition explains both findings. Innovation uncertainty operates through variation in continuation values within destinations, while transition uncertainty operates through variation in expected continuation values across destinations. Both these exposures and the separately calibrated penalty parameters contribute to the relative size of the two effects.

In Sargent et al. (2026a), participation in the U.S. tax-transfer-and-insurance system is worth about four percentage points of annual growth to a household at the 10th percentile and costs a household at the 90th percentile more than one percentage point. Eliminating consumption risk at a fixed percentile is worth about one tenth of a percentage point. At $\bar{p} = 0.45$, the joint cost of consumption risk remains an order of magnitude below that insurance benefit. At the lowest target, $\bar{p} = 0.10$, the joint cost rises to 1.3–1.5 annual percentage points, putting it in the same order of magnitude as the insurance benefit at the bottom of the distribution.

Sargent et al. (2026a) also help explain the size of these increments. They use an orthogonal shock decomposition and find that permanent, martingale risk accounts for 94–98 percent of the compensating difference. Our decomposition in Section 5.1 separates innovations by their arrival date. We find that the persistent component is about 98 percent of the Case 0 cost under consumption ranking and 79–80 percent under private income. A distortion of consumption innovations affects the whole future path of log consumption, amplifying its welfare cost under multiplier preferences.

Three extensions seem worth pursuing. The first is to let mobility itself vary with observables that we have held fixed here. Transition probabilities plausibly depend on age, on cohort, on household composition, and on aggregate conditions. We could allow for these differences while retaining a common law of motion for households with the same characteristics and percentile. The second is to study both channels in an equilibrium model. We have taken the percentile process as given and measured its law of motion in the data, so the welfare costs we report are those of an endowment household that cannot alter its own exposure. Placing the nested recursion inside a model with saving, labor supply, and endogenous mobility would make consumption dynamics and the transition probabilities equilibrium objects, and would allow us to study how household choices and public policy interact with innovation uncertainty and with transition uncertainty. The third is to trace the full set of

penalty pairs $(\theta_\varepsilon, \theta_\Pi)$ that attain a common detection-error target. We could then compare welfare costs as concern shifts between the two sources, extending our separate calibrations and common-penalty joint case.

The appendices are organized as follows. Appendix A collects proofs of the formal results stated in the text, and Appendix B gives direct likelihood-ratio derivations of Cases 1–3. Appendices C–D relate the model to established formulas and derive the welfare-path loadings. Appendix E establishes convergence of value-function iteration. Appendix F records the data construction. Appendices G–H document the state reduction and computation. Appendix I reports retained variance and forecast accuracy across DMD truncation ranks, and Appendix J compares the joint and separately calibrated welfare increments.

A Proofs of stated results

A.1 Exponential tilting under relative entropy

Proof. We fix (p, ζ, c) and write $Y(j, e) := \beta v_j(\zeta', c')$ for the outcome $(p', \varepsilon) = (j, e)$. We then form the Lagrangian, with multiplier ς on the mean-one constraint:

$$\mathcal{L}(h, \varsigma) = \mathbb{E}_Q[h(Y + \theta \log h) \mid p, \zeta, c] + \varsigma (\mathbb{E}_Q[h \mid p, \zeta, c] - 1). \quad (74)$$

Since $d(h \log h)/dh = 1 + \log h$, the first-order condition on the support of Q is

$$Y + \theta(1 + \log h) + \varsigma = 0, \quad h = K \exp(-Y/\theta), \quad K := \exp(-1 - \varsigma/\theta) > 0.$$

The mean-one constraint determines the normalizing constant:

$$K^{-1} = \mathbb{E}_Q[\exp(-Y/\theta) \mid p, \zeta, c].$$

Substituting $Y = \beta v_{p'}(\zeta', c')$ gives (11).

For the minimized value, set $Z := \mathbb{E}_Q[\exp(-Y/\theta) \mid p, \zeta, c]$. Direct substitution yields

$$\begin{aligned} \mathbb{E}_Q[h^*(Y + \theta \log h^*) \mid p, \zeta, c] &= \mathbb{E}_Q \left[\frac{e^{-Y/\theta}}{Z} \left(Y + \theta \log \frac{e^{-Y/\theta}}{Z} \right) \mid p, \zeta, c \right] \\ &= \mathbb{E}_Q \left[\frac{e^{-Y/\theta}}{Z} (-\theta \log Z) \mid p, \zeta, c \right] = -\theta \log Z, \end{aligned}$$

which proves (12). The candidate likelihood ratio is positive on the support of Q and satisfies the mean-one constraint. Because $x \mapsto x \log x$ is strictly convex on $[0, \infty)$, the objective is strictly convex over feasible likelihood ratios. The candidate is therefore the unique minimizer up to Q -a.s. equivalence. $\square$

A.2 The two-stage recursion

Proof. For a fixed transition vector $\tilde{\Pi}_{p\cdot}$, the objective (15) separates across the conditional innovation densities. We therefore minimize first over $\{\tilde{\varphi}_j\}$. For each j in the support of $\Pi_{p\cdot}$, the inner problem is

$$\min_{\tilde{\varphi}_j} \int \beta v_j(\zeta', c') \tilde{\varphi}_j \, d\varepsilon + \theta_\varepsilon \text{ent}(\tilde{\varphi}_j \| \varphi),$$

which is (10) with $\mathbb{E}_Q$ replaced by $\mathbb{E}_\varphi$ and θ by θ_ε . Lemma 1 therefore gives the minimized value $(\mathcal{R}_\varepsilon v_j)(\zeta, c)$ and minimizer (20). For a destination with $\Pi_{pj} = 0$, the conditional innovation density does not affect the objective, and we may set it equal to φ . Substituting reduces the outer problem to

$$\min_{\tilde{\Pi}_{p\cdot}} \sum_j \tilde{\Pi}_{pj} (\mathcal{R}_\varepsilon v_j)(\zeta, c) + \theta_\Pi \text{ent}(\tilde{\Pi}_{p\cdot} \| \Pi_{p\cdot}),$$

which is again an instance of (10), now over the finite set $\{0, \dots, M-1\}$ with reference measure $\Pi_{p\cdot}$, payoff $(\mathcal{R}_\varepsilon v_j)$, and penalty θ_Π . Lemma 1 delivers (17) and (19). $\square$

A.3 The local expansion

Proof. For a random variable X with a cumulant-generating function that is finite in a neighborhood of the origin,

$$\log \mathbb{E}[e^{tX}] = t \mathbb{E}[X] + \frac{t^2}{2} \text{Var}(X) + O(t^3) \quad \text{as } t \rightarrow 0.$$

We set $t = -1/\theta$ to obtain the entropic expansion

$$-\theta \log \mathbb{E}[e^{-X/\theta}] = \mathbb{E}[X] - \frac{1}{2\theta} \text{Var}(X) + O(\theta^{-2}). \quad (75)$$

We apply (75) to the inner stage (16), with $X = Y_j$ and expectation over ε :

$$(\mathcal{R}_\varepsilon v_j)(\zeta, c) = \mathbb{E}_\varepsilon[Y_j] - \frac{1}{2\theta_\varepsilon} \text{Var}_\varepsilon(Y_j) + O(\theta_\varepsilon^{-2}).$$

Because the set of next-period percentiles is finite, we can choose the remainder uniformly over $j \in \text{supp}(\Pi_{p\cdot})$. Applying (75) to the outer stage (17), with $X = (\mathcal{R}_\varepsilon v_j)(\zeta, c)$ and expectation over $j \sim \Pi_{p\cdot}$, gives

$$\mathcal{T}_{\theta_\varepsilon, \theta_\Pi}[\beta v](p, \zeta, c) = \mathbb{E}_{\Pi_p}[\mathcal{R}_\varepsilon v_j] - \frac{1}{2\theta_\Pi} \text{Var}_{\Pi_p}(\mathcal{R}_\varepsilon v_j) + O(\theta_\Pi^{-2}).$$

We substitute the inner expansion into the outer mean to obtain

$$\mathbb{E}_{\Pi_p}[\mathbb{E}_\varepsilon[Y_j]] - \frac{1}{2\theta_\varepsilon}\mathbb{E}_{\Pi_p}[\text{Var}_\varepsilon(Y_j)] + O(\theta_\varepsilon^{-2}).$$

Moreover, the inner expansion implies

$$\text{Var}_{\Pi_p}(\mathcal{R}_\varepsilon v_j) = \text{Var}_{\Pi_p}(\mathbb{E}_\varepsilon[Y_j]) + O(\theta_\varepsilon^{-1}).$$

The approximation error in the outer variance therefore contributes $O((\theta_\varepsilon\theta_\Pi)^{-1})$. Combining these terms gives (22). $\square$

A.4 Affine value functions for Cases 0 and 1

Proof. We look for an affine fixed point of the form $v_j(\zeta, c) = \lambda_j^\top \zeta + b_j^\top c + \iota_j$. The state dynamics imply

$$v_j(\zeta', c') = \underbrace{\iota_j + b_j^\top c + (\lambda_j^\top A + b_j^\top D)\zeta}_{=: \mu_j(\zeta, c)} + \underbrace{(\lambda_j^\top C + b_j^\top F)\varepsilon}_{=: s_j^\top \varepsilon}. \quad (76)$$

Since $\mathbb{E}_\varphi[\varepsilon] = 0$, substitution into the Case 0 recursion gives

$$\begin{aligned} v_p^{(0)}(\zeta, c) &= (1 - \beta)e_p^\top c + \beta \sum_j \Pi_{pj} \mu_j(\zeta, c) \\ &= (1 - \beta)e_p^\top c + \beta \sum_j \Pi_{pj} (\iota_j + b_j^\top c + (\lambda_j^\top A + b_j^\top D)\zeta). \end{aligned}$$

We match the coefficients of c and ζ to obtain (34)–(35). The representation

$$W = (1 - \beta) \sum_{h=0}^{\infty} \beta^h \Pi^h$$

shows that W has nonnegative entries and that each row sums to one. The stated nonsingularity condition gives a unique solution to the Sylvester equation for Λ .

If ζ contains a constant coordinate, its loading is not separately identified from the intercept, so we impose the stated normalization. Otherwise, matching constants gives $\iota = \beta \Pi \iota$ in Case 0 and hence $\iota = 0$ because $\rho(\beta \Pi) = \beta < 1$.

Because $s_j^\top \varepsilon \sim N(0, \mathbf{e}_j)$, the Gaussian moment formula applied to (76) gives

$$-\theta \log \mathbb{E}_\varphi \left[\exp \left(-\frac{\beta}{\theta} v_j(\zeta', c') \right) \mid \zeta, c \right] = \beta \mu_j(\zeta, c) - \frac{\beta^2}{2\theta} \mathbf{e}_j. \quad (77)$$

We substitute this expression into the Case 1 recursion. The slope equations do not change, while the intercept satisfies $(I - \beta\Pi)\iota = -(\beta^2/(2\theta))\Pi\mathbf{e}$. We set $\kappa := \iota$ to obtain (36). Its entries are nonpositive because $(I - \beta\Pi)^{-1}\Pi$ has non-negative entries.

Finally, Lemma 1 and (76) give the unnormalized likelihood ratio

$$g_j^*(\varepsilon) \propto \exp\left(-\frac{\beta}{\theta}s_j^\top\varepsilon\right).$$

Because $s_j^\top\varepsilon \sim N(0, \mathbf{e}_j)$, the Gaussian moment-generating function gives

$$\mathbb{E}_\varphi\left[\exp\left(-\frac{\beta}{\theta}s_j^\top\varepsilon\right)\right] = \exp\left(\frac{\beta^2}{2\theta^2}\mathbf{e}_j\right).$$

The normalized likelihood ratio is therefore

$$g_j^*(\varepsilon) = \exp\left(-\frac{\beta}{\theta}s_j^\top\varepsilon - \frac{\beta^2}{2\theta^2}\mathbf{e}_j\right). \quad (78)$$

For $P = N(\mathbf{m}, I_k)$ and reference density $\varphi = N(0, I_k)$, taking the ratio of their densities gives

$$\frac{dP}{d\varphi}(\varepsilon) = \exp\left(\mathbf{m}^\top\varepsilon - \frac{1}{2}\|\mathbf{m}\|^2\right).$$

We set $\mathbf{m} = -(\beta/\theta)s_j$, which reproduces (78). Thus the worst-case innovation distribution is $N(-(\beta/\theta)s_j, I_k)$.

The affine functions constructed above satisfy the Case 0 and Case 1 Bellman equations and belong to the linear-growth class in Proposition 13. That proposition establishes uniqueness of the Bellman fixed point in this class, so these affine fixed points are the two value functions stated in the proposition. $\square$

A.5 Translation under a common consumption shift

Proof. For any random variable X , scalar b , and $\vartheta \in (0, \infty]$, the certainty-equivalent operator satisfies

$$\mathcal{E}_\vartheta(X + b) = \begin{cases} -\vartheta \log \mathbb{E}[e^{-(X+b)/\vartheta}], & \vartheta < \infty, \\ \mathbb{E}[X + b], & \vartheta = \infty, \end{cases} = \mathcal{E}_\vartheta(X) + b.$$

Thus both stages of (18) are translation equivariant.

We first establish the translation property for the finite-horizon iterates. We set $v^{[0]} \equiv 0$

and $v^{[i+1]} = \Phi v^{[i]}$, where Φ is the Bellman operator in (18). We claim

$$v_p^{[i]}(\zeta, c + a\mathbf{1}) = v_p^{[i]}(\zeta, c) + (1 - \beta^i)a. \quad (79)$$

The claim holds for $i = 0$. Suppose it holds for i . Starting from $c + a\mathbf{1}$ shifts next period's consumption vector by the same $a\mathbf{1}$. By the induction hypothesis and translation equivariance, the nested continuation term therefore increases by $\beta(1 - \beta^i)a$. The current payoff increases by $(1 - \beta)a$, so

$$(1 - \beta)a + \beta(1 - \beta^i)a = (1 - \beta^{i+1})a.$$

Proposition 13 implies that these iterates converge pointwise to the infinite-horizon value function. Taking the limit in (79) proves the first equality in (39). For w_p , a common shift raises every $\bar{c}_{j,h}$ in (38) by a . Since every row of Π^h sums to one,

$$w_p(\zeta, c + a\mathbf{1}) - w_p(\zeta, c) = (1 - \beta) \sum_{h=0}^{\infty} \beta^h a = a.$$

□

A.6 Closed-form compensating differences for Cases 0 and 1

Proof. We substitute $\bar{c}_{j,h} = \mu_{j,h} + \frac{1}{2}\sigma_{j,h}^2$ from (37) into (38). The terms involving $\mu_{j,h}$ equal the Case 0 value because Case 0 evaluates expected log consumption under the approximating model. The remaining terms are the discounted conditional variances. Since $\sigma_{j,0}^2 = 0$, these terms begin at horizon one, giving

$$w_p(\zeta, c) = v_p^{(0)}(\zeta, c) + \frac{1 - \beta}{2} \sum_{h=1}^{\infty} \beta^h [\Pi^h \sigma_h^2]_p. \quad (80)$$

Using (41), we subtract the Case 0 value from (80) to obtain (42). Since $v_p^{(1)} = v_p^{(0)} + \kappa_p$, we also have $\alpha_p^{(1)} = \alpha_p^{(0)} - \kappa_p$, which gives (43). For $\Pi = I$,

$$\begin{aligned} \alpha_p^{(0)} &= \frac{1 - \beta}{2} \sum_{h \geq 1} \beta^h \sum_{i=0}^{h-1} \|J_{p,i}\|^2 \\ &= \frac{1 - \beta}{2} \sum_{i \geq 0} \|J_{p,i}\|^2 \sum_{h > i} \beta^h = \frac{\beta}{2} \sum_{i \geq 0} \beta^i \|J_{p,i}\|^2. \end{aligned}$$

□

A.7 Mobility and the stationary-weighted mean

Proof. We multiply (42) on the left by $\pi^\top$. Stationarity gives $\pi^\top \Pi^h = \pi^\top$ for every $h \geq 1$, so

$$\pi^\top \alpha^{(0)}(\Pi) = \frac{1-\beta}{2} \sum_{h=1}^{\infty} \beta^h \pi^\top \sigma_h^2.$$

The vectors σ_h^2 depend on (A, C, D, F) through (90) and do not depend on Π . Setting $\Pi = I$ therefore gives the same expression. Because every probability vector is stationary for I , we may use the same weights π on both sides. $\square$

B Detailed derivations of Cases 1–3

B.1 Case 1: innovation uncertainty

Case 1 follows by sending θ_Π to infinity in Proposition 2. To derive it directly, fix (p, ζ, c) , suppress this dependence in the notation, and restrict the likelihood ratio to

$$h(p', \varepsilon) = g_{p'}(\varepsilon), \quad \mathbb{E}_\varphi[g_j(\varepsilon) \mid \zeta, c] = 1 \quad \text{for every } j. \quad (81)$$

The separate normalizations keep the transition probabilities fixed.

Expanding the conditional expectation in (8) under the kernel (5), and substituting $h(j, \varepsilon) = g_j(\varepsilon)$, gives

$$\sum_{j=0}^{M-1} \Pi_{pj} \mathbb{E}_\varphi[g_j(\varepsilon) (\beta v_j(\zeta', c') + \theta \log g_j(\varepsilon)) \mid \zeta, c]. \quad (82)$$

Under the restrictions in (81), minimization separates over j :

$$\min_{\substack{g_j \geq 0 \\ \mathbb{E}_\varphi[g_j \mid \zeta, c] = 1}} \mathbb{E}_\varphi[g_j(\varepsilon) (\beta v_j(\zeta', c') + \theta \log g_j(\varepsilon)) \mid \zeta, c]. \quad (83)$$

For each j in the support of $\Pi_{p\cdot}$, we apply Lemma 1 with $Y_j := \beta v_j(\zeta', c')$ and reference distribution φ . The minimizing conditional likelihood ratio is

$$g_j^*(\varepsilon) = \frac{\exp(-Y_j/\theta)}{\mathcal{Z}_j}, \quad \mathcal{Z}_j = \mathbb{E}_\varphi[\exp(-Y_j/\theta) \mid \zeta, c],$$

and the minimized conditional value is $-\theta \log \mathcal{Z}_j$. The minimizing likelihood ratio agrees with (27). We weight the minimized values by Π_{pj} in (82) and add the current payoff to

obtain (26).

B.2 Case 2: transition uncertainty

We fix (p, ζ, c) , suppress this dependence in the notation, and restrict the Case 2 likelihood ratio to

$$h(p', \varepsilon) = \varrho(p'), \quad \sum_{j=0}^{M-1} \Pi_{pj} \varrho(j) = 1. \quad (84)$$

Because ϱ is independent of ε , the innovation distribution remains $N(0, I_k)$.

We take the innovation expectation first and use (28). The continuation problem is

$$\min_{\substack{\varrho(j) \geq 0 \\ \sum_j \Pi_{pj} \varrho(j) = 1}} \sum_{j=0}^{M-1} \Pi_{pj} \varrho(j) (\beta \bar{v}_j(\zeta, c) + \theta \log \varrho(j)). \quad (85)$$

We apply Lemma 1 to the finite reference distribution Π_p and payoff $\beta \bar{v}_j(\zeta, c)$. On the support of the reference row, the minimizing transition likelihood ratio is

$$\varrho^*(j) = \frac{\exp[-\beta \bar{v}_j(\zeta, c)/\theta]}{\sum_k \Pi_{pk} \exp[-\beta \bar{v}_k(\zeta, c)/\theta]}.$$

Because $\tilde{\Pi}_{pj} = \Pi_{pj} \varrho^*(j)$, we obtain (30). The minimized continuation value is

$$-\theta \log \sum_j \Pi_{pj} \exp[-\beta \bar{v}_j(\zeta, c)/\theta].$$

We add the current payoff to obtain (29).

B.3 Case 3: joint uncertainty

In Case 3, any nonnegative joint likelihood ratio satisfying the mean-one constraint is admissible. Lemma 1 therefore applies directly. Under kernel (5),

$$\mathbb{E}_Q \left[\exp \left(-\frac{\beta}{\theta} v_{p'}(\zeta', c') \right) \middle| p, \zeta, c \right] = \sum_{j=0}^{M-1} \Pi_{pj} \mathcal{M}_j(\zeta, c).$$

We substitute this expression into (12) to obtain (32). We also substitute it into (11) to obtain the minimizing joint likelihood ratio

$$h^*(j, \varepsilon \mid p, \zeta, c) = \frac{\exp[-(\beta/\theta) v_j(\zeta', c')]}{\sum_k \Pi_{pk} \mathcal{M}_k(\zeta, c)}.$$

We integrate this joint likelihood ratio over the reference innovation distribution, conditional on $p' = j$, to obtain the transition likelihood ratio in (19). We then divide the joint likelihood ratio by the transition likelihood ratio to obtain the conditional innovation likelihood ratio in (20). The alternative joint density with respect to counting measure and Lebesgue measure is $\Pi_{pj}\varphi(\varepsilon)h^*(j, \varepsilon \mid p, \zeta, c)$.

C Special cases and recursive-utility mappings

Remark 8 (Relation to Hansen and Sargent (2008a)). When $M = 1$, $\Pi = [1]$, and (34) gives $W = 1$. Therefore $s_0 = C^\top \lambda + F_0^\top$, and (36) becomes

$$\kappa = -\frac{\beta^2}{2(1-\beta)\theta} \|C^\top \lambda + F_0^\top\|^2.$$

This is the scalar multiplier-preference formula of Hansen and Sargent (2008a). Proposition 5 replaces its scalar resolvent $(1-\beta)^{-1}$ with the Markov resolvent $(I-\beta\Pi)^{-1}$.

Remark 9 (Equal conditional innovation variance). If $\mathbf{e}_j = \bar{\mathbf{e}}$ for every j , then

$$\alpha^{(1)} - \alpha^{(0)} = -\kappa = \frac{\beta^2 \bar{\mathbf{e}}}{2\theta(1-\beta)} \mathbf{1}.$$

For a fixed common variance $\bar{\mathbf{e}}$, the increment is the same at every percentile and for every transition matrix Π , since $(I-\beta\Pi)^{-1}\Pi\mathbf{1} = (1-\beta)^{-1}\mathbf{1}$. The common variance can itself change with Π through the continuation-value loadings s_j .

Remark 10 (Mapping to Epstein–Zin preferences). When $\theta_\varepsilon = \theta_\Pi = \theta$, the two-stage operator reduces to one entropic certainty equivalent under the joint kernel Q . We can therefore write the Case 3 Bellman equation (32) as

$$v_p = (1-\beta)c_p + \beta \left[-\frac{\theta}{\beta} \log \mathbb{E}_Q \exp \left(-\frac{\beta}{\theta} v_{p'} \right) \right].$$

With an intertemporal elasticity of substitution equal to one, the logarithmic Epstein–Zin recursion is

$$v_p = (1-\beta)c_p + \frac{\beta}{1-\gamma} \log \mathbb{E}_Q \exp((1-\gamma)v_{p'}).$$

Matching the exponent and the coefficient gives

$$\gamma = 1 + \frac{\beta}{\theta}. \tag{86}$$

Thus, equal penalties make Case 3 equivalent to a unit-elasticity Epstein–Zin recursion with relative risk aversion γ . In Case 1, the Epstein–Zin mapping applies to the innovation aggregator conditional on the destination percentile, with curvature coefficient $1 + \beta/\theta_\varepsilon$. We then average these conditional values using Π .

Remark 11 (Fixed percentiles and a consumption random walk). Setting $\Pi = I$ in (42) gives

$$\alpha_p^{(0)} = \frac{\beta}{2} \sum_{i \geq 0} \beta^i \|J_{p,i}\|^2,$$

which is equation (35) in Sargent et al. (2026a). If, in addition, $M = 1$, $D = 0$, and $\|F_0\|^2 = \sigma_\varepsilon^2$, consumption follows a scalar random walk. In this case, $J_{0,i} = F_0$ for every i , and

$$\alpha^{(0)} = \frac{\beta}{2(1-\beta)} \sigma_\varepsilon^2,$$

while

$$\frac{\alpha^{(1)}}{\alpha^{(0)}} = 1 + \frac{\beta}{\theta_\varepsilon}. \quad (87)$$

These are the Obstfeld–Tallarini compensating difference and its linear scaling with relative risk aversion (Obstfeld, 1994, Tallarini, 2000).

The stable dynamics (A, C, D) can give i.i.d. innovations persistent effects on consumption growth. With serially correlated growth, (87) need not hold: $\mathbf{e}_p$ uses the innovation's discounted loading on the continuation value, whereas $\sigma_{p,h}^2$ uses its horizon-specific loading on consumption.

D Welfare-path calculations

For $r \geq 0$, repeated substitution in (1) gives

$$\zeta_{t+r} = A^r \zeta_t + \sum_{\ell=1}^r A^{r-\ell} C \varepsilon_{t+\ell},$$

where the sum is empty when $r = 0$. We then sum (2) from date t through date $t + h - 1$:

$$c_{t+h} = c_t + \sum_{r=0}^{h-1} D \zeta_{t+r} + \sum_{\ell=1}^h F \varepsilon_{t+\ell}.$$

We substitute the first expression into the second and collect the loading on each innovation. For $h \geq 1$, this gives

$$c_{p,t+h} = c_{p,t} + \psi_{p,h}^\top \zeta_t + \sum_{i=0}^{h-1} J_{p,i} \varepsilon_{t+h-i}, \quad (88)$$

where

$$\psi_{p,h}^\top = \sum_{s=0}^{h-1} D_p A^s, \quad (89)$$

$$J_{p,i} = F_p + \sum_{r=0}^{i-1} D_p A^r C, \quad i \geq 0. \quad (90)$$

Because the innovations are independent standard normal vectors,

$$\mu_{p,h}(\zeta_t, c_t) = c_{p,t} + \psi_{p,h}^\top \zeta_t, \quad \sigma_{p,h}^2 = \sum_{i=0}^{h-1} \|J_{p,i}\|^2. \quad (91)$$

For the fitted full-state system, $\zeta_t = (1, y_t^\top)^\top$ and $A = \text{diag}(1, B)$ by (54). The leading coordinate generates deterministic growth and has zero loading in $A^r C$ because the corresponding row of C is zero. The stochastic innovation loadings are therefore bounded when $\rho(B) < 1$. Under the reduced representation, the analogous condition is $\rho(\tilde{B}) < 1$. Under either condition, $J_{p,i}$ is bounded, $\sigma_{p,h}^2$ grows at most linearly with h , and the discount factor β^h makes the welfare series in Section 3.1.1 finite.

E Existence, uniqueness, and value-function iteration

Let Φ denote the Bellman operator in (18), and stack the continuous state as $z := (\zeta^\top, c^\top)^\top$. Its law is

$$z' = Gz + H\varepsilon, \quad G := \begin{bmatrix} A & 0 \\ D & I_M \end{bmatrix}, \quad H := \begin{bmatrix} C \\ F \end{bmatrix}, \quad \varepsilon \sim N(0, I_k). \quad (92)$$

The payoff is $r_p(z) := (1 - \beta)c_p$. Because G is block lower triangular, $\rho(G) = \max\{\rho(A), 1\}$. In the fitted full-state system, $A = \text{diag}(1, B)$ and $\rho(B) < 1$, so $\rho(G) = 1 < 1/\beta$. The proposition below uses the more general condition $\rho(G) < 1/\beta$.

Both the state and the payoff are unbounded, so the contraction theorem for bounded value functions does not apply directly. This problem is studied by Christensen (2022) for recursive utility, including robust preferences in linear-Gaussian environments. Weighted-

supremum methods for dynamic programs with unbounded rewards are developed by Lippman (1975) and Wessels (1977). Related risk-sensitive problems are studied by Shen et al. (2013) and Bäuerle and Jaśkiewicz (2018). We prove the result by constructing lower and upper solutions and applying an order-interval fixed-point argument related to Du (1990).

It is convenient to write both certainty equivalents in (18) under one convention. For a random variable X and a penalty $\vartheta \in (0, \infty]$, set

$$\mathcal{E}_\vartheta(X) := \begin{cases} -\vartheta \log \mathbb{E}[\exp(-X/\vartheta)], & \vartheta < \infty, \\ \mathbb{E}[X], & \vartheta = \infty. \end{cases} \quad (93)$$

Each $\mathcal{E}_\vartheta$ is increasing, concave, and translation equivariant, so that $\mathcal{E}_\vartheta(X + b) = \mathcal{E}_\vartheta(X) + b$ for every constant b , and Jensen's inequality gives $\mathcal{E}_\vartheta(X) \leq \mathbb{E}[X]$. The inner operator (16) is concave in the next-period value function, and the outer operator in (18) is concave and increasing in each next-period percentile's inner certainty equivalent. The composition inherits both properties, so Φ is an increasing and concave operator.

We first state an order-interval lemma that identifies the properties of the Bellman operator needed for convergence.⁸

Lemma 12 (Geometric convergence on an order interval). *Let $w \geq 1$ be a weight function, let*

$$\|v\|_w := \max_p \sup_z \frac{|v_p(z)|}{w(z)},$$

and let $\underline{v} \leq \bar{v}$ be M -vector functions with finite $\|\cdot\|_w$ norm. Let T be an increasing and concave self-map of the order interval $[\underline{v}, \bar{v}]$ such that, for some $\eta \in (0, 1)$,

$$T\underline{v} \geq \underline{v} + \eta(\bar{v} - \underline{v}), \quad T\bar{v} \leq \bar{v}. \quad (94)$$

Then T has a unique fixed point v^ in $[\underline{v}, \bar{v}]$, and every $v^{[0]} \in [\underline{v}, \bar{v}]$ satisfies*

$$\|T^n v^{[0]} - v^*\|_w \leq (1 - \eta)^n \|\bar{v} - \underline{v}\|_w, \quad n \geq 0. \quad (95)$$

Proof. We set $\underline{v}_0 := \underline{v}$ and $\bar{v}_0 := \bar{v}$ and iterate $\underline{v}_{n+1} := T\underline{v}_n$ and $\bar{v}_{n+1} := T\bar{v}_n$. The endpoint inequalities (94) and monotonicity give

$$\underline{v}_0 \leq \underline{v}_1 \leq \dots \leq \bar{v}_1 \leq \bar{v}_0. \quad (96)$$

⁸The argument is a direct specialization of fixed-point results for increasing concave operators on order intervals, such as Du (1990, Theorem 3.1).

We show by induction that

$$\underline{v}_{n+1} \geq (1 - \eta)\underline{v}_n + \eta \bar{v}_n \quad \text{for every } n \geq 0. \quad (97)$$

The case $n = 0$ is the first inequality in (94). If (97) holds at n , then monotonicity and concavity give

$$\underline{v}_{n+2} = T\underline{v}_{n+1} \geq T((1 - \eta)\underline{v}_n + \eta \bar{v}_n) \geq (1 - \eta)T\underline{v}_n + \eta T\bar{v}_n = (1 - \eta)\underline{v}_{n+1} + \eta \bar{v}_{n+1},$$

which is (97) at $n + 1$. Combining (96) and (97),

$$0 \leq \bar{v}_{n+1} - \underline{v}_{n+1} \leq \bar{v}_n - [(1 - \eta)\underline{v}_n + \eta \bar{v}_n] = (1 - \eta)(\bar{v}_n - \underline{v}_n),$$

so that

$$\|\bar{v}_n - \underline{v}_n\|_w \leq (1 - \eta)^n \|\bar{v} - \underline{v}\|_w. \quad (98)$$

By (96), $\underline{v}_n$ increases pointwise and $\bar{v}_n$ decreases pointwise. Equation (98) shows that the distance between them converges to zero in the weighted norm. They therefore have a common pointwise limit v^* , with $\underline{v}_n \leq v^* \leq \bar{v}_n$ for every n . Monotonicity gives

$$\underline{v}_{n+1} = T\underline{v}_n \leq Tv^* \leq T\bar{v}_n = \bar{v}_{n+1},$$

and taking pointwise limits in these bounds yields $Tv^* = v^*$.

If $v^{[0]} \in [\underline{v}, \bar{v}]$, then monotonicity gives $\underline{v}_n \leq T^n v^{[0]} \leq \bar{v}_n$, and (98) implies (95). Applying (95) to a second fixed point $\tilde{v}$ in the interval gives $\|\tilde{v} - v^*\|_w \leq (1 - \eta)^n \|\bar{v} - \underline{v}\|_w$ for every n , hence $\tilde{v} = v^*$. $\square$

Let $\mathcal{V}_1$ denote the class of measurable functions $v : \{0, \dots, M-1\} \times \mathbb{R}^{n+M} \rightarrow \mathbb{R}$ satisfying

$$\|v\|_{\mathcal{V}_1} := \max_{p=0, \dots, M-1} \sup_{z \in \mathbb{R}^{n+M}} \frac{|v_p(z)|}{1 + \|z\|_2} < \infty. \quad (99)$$

Thus a function belongs to $\mathcal{V}_1$ when its absolute value is bounded by a constant times $1 + \|z\|_2$. This class includes affine and nonlinear functions. The choice of norm in (99) is immaterial because all norms on $\mathbb{R}^{n+M}$ are equivalent.

Proposition 13 (Existence, uniqueness, and convergence from every linear-growth initial value). *Suppose $\rho(G) < 1/\beta$ and fix $(\theta_\varepsilon, \theta_\Pi) \in (0, \infty]^2$. Then Φ is a well-defined self-map of $\mathcal{V}_1$ and has a unique fixed point $v^* \in \mathcal{V}_1$. For every initial value $v^{[0]} \in \mathcal{V}_1$ there are a norm $\|\cdot\|_*$ on $\mathbb{R}^{n+M}$, a constant $a > 0$, the weight $w(z) = a + \|z\|_*$, and constants $C_0 < \infty$ and*

$q \in (0, 1)$ such that

$$\|\Phi^i v^{[0]} - v^*\|_w \leq C_0 q^i, \quad i \geq 0. \quad (100)$$

Proof. Because $\rho(G) < 1/\beta$, Horn and Johnson (2012, Lemma 5.6.10) supplies an equivalent vector norm $\|\cdot\|_*$ whose induced matrix norm satisfies

$$g := \|G\|_* < \frac{1}{\beta}. \quad (101)$$

We fix $K < \infty$ such that $\max_p |r_p(z)| \leq K\|z\|_*$ and define

$$\sigma := \|H\|_{2 \rightarrow *}, \quad \bar{m} := \mathbb{E}\|\varepsilon\|_2.$$

For an initial value satisfying $\max_p |v_p^{[0]}(z)| \leq C_1(1 + \|z\|_*)$ for some $C_1 < \infty$, we choose

$$R > \max \left\{ C_1, \frac{K}{1 - \beta g} \right\} \quad (102)$$

and define

$$\Xi_R := \begin{cases} \theta_\varepsilon \log \mathbb{E} \exp \left(\frac{\beta R \sigma}{\theta_\varepsilon} \|\varepsilon\|_2 \right), & \theta_\varepsilon < \infty, \\ \beta R \sigma \bar{m}, & \theta_\varepsilon = \infty. \end{cases} \quad (103)$$

Gaussian exponential moments make Ξ_R finite, and Jensen's inequality gives $\Xi_R \geq \beta R \sigma \bar{m}$.

We choose

$$a > \max \left\{ 1, \frac{\Xi_R}{R(1 - \beta)} \right\}, \quad w(z) := a + \|z\|_*, \quad (104)$$

and define the lower and upper bounding functions

$$\underline{v}_p(z) := -Rw(z), \quad \bar{v}_p(z) := Rw(z), \quad p = 0, \dots, M - 1. \quad (105)$$

Then $v^{[0]} \in [\underline{v}, \bar{v}]$. We write $z' := Gz + H\varepsilon$ for the next state. It follows that

$$w(z') \leq a + g\|z\|_* + \sigma\|\varepsilon\|_2, \quad (106)$$

and the finiteness of Ξ_R also ensures that Φ is well defined on $[\underline{v}, \bar{v}]$.

Both bounding functions assign the same value to every next-period percentile. The inner certainty equivalent is therefore common across next-period percentiles, and the outer certainty equivalent returns it unchanged. For the upper bounding function, $\mathcal{E}_\theta(X) \leq \mathbb{E}[X]$

and (106) give

$$(\Phi\bar{v})_p(z) \leq K\|z\|_* + \beta R(a + g\|z\|_* + \sigma\bar{m}) \leq R(a + \|z\|_*) = \bar{v}_p(z), \quad (107)$$

where we use $K + \beta Rg < R$ from (102) together with $\beta Ra + \beta R\sigma\bar{m} \leq \beta Ra + \Xi_R < Ra$ from (104). For the lower bounding function and $\theta_\varepsilon < \infty$, (106) and the definition of Ξ_R give

$$\mathcal{E}_{\theta_\varepsilon}(-\beta R w(z')) = -\theta_\varepsilon \log \mathbb{E} \exp\left(\frac{\beta R}{\theta_\varepsilon} w(z')\right) \geq -\beta R(a + g\|z\|_*) - \Xi_R,$$

and the same bound follows directly from the expectation when $\theta_\varepsilon = \infty$. Subtracting $\underline{v}_p(z) = -R(a + \|z\|_*)$,

$$(\Phi\underline{v})_p(z) - \underline{v}_p(z) \geq [R(1 - \beta)a - \Xi_R] + [R(1 - \beta g) - K]\|z\|_*. \quad (108)$$

The two numbers

$$\delta_0 := 1 - \beta - \frac{\Xi_R}{Ra}, \quad \delta_1 := 1 - \beta g - \frac{K}{R} \quad (109)$$

are strictly positive by (102) and (104). We set $\eta := \frac{1}{2} \min\{\delta_0, \delta_1\} \in (0, 1)$. The right-hand side of (108) equals $Ra\delta_0 + R\delta_1\|z\|_*$, while $\bar{v}_p(z) - \underline{v}_p(z) = 2R(a + \|z\|_*)$. By the definition of η , the first expression is at least $\eta(\bar{v}_p - \underline{v}_p)(z)$. Together with (107),

$$\Phi\underline{v} \geq \underline{v} + \eta(\bar{v} - \underline{v}), \quad \Phi\bar{v} \leq \bar{v}.$$

These are the hypotheses of Lemma 12, and $\|\bar{v} - \underline{v}\|_w = 2R$. Hence Φ has a unique fixed point in $[\underline{v}, \bar{v}]$ and

$$\|\Phi^i v^{[0]} - v^*\|_w \leq 2R(1 - \eta)^i, \quad (110)$$

which is (100) with $C_0 = 2R$ and $q = 1 - \eta$.

For uniqueness over $\mathcal{V}_1$, we take two fixed points v^* and $\tilde{v}$ in $\mathcal{V}_1$. We choose R above both growth bounds and then choose a as in (104). Both functions lie in the resulting common interval, whose fixed point is unique, so $v^* = \tilde{v}$. Every initialization in $\mathcal{V}_1$ can be placed in such an interval, so value-function iteration converges geometrically to the same fixed point from any starting point in $\mathcal{V}_1$. Because the initialization was arbitrary, the same construction also shows that every $v \in \mathcal{V}_1$ is mapped to a finite-valued element of $\mathcal{V}_1$. Thus Φ is a well-defined self-map of this class. $\square$

The spectral-radius and exponential-moment assumptions enter different parts of the proof. The inequality $\beta g < 1$ controls how the current state propagates through the linear dynamics. The Gaussian exponential moment controls the constant term through Ξ_R .

F CEX data construction

We construct the CEX variables following Sargent et al. (2026a). Within each specification, we use one common ranking variable to assign percentile groups. Under private-income ranking, private income, post-tax income, and consumption are all grouped by private income. Under consumption ranking, they are all grouped by consumption. This differs from Sargent et al. (2026a), who rank each variable by its own distribution.

Before 2015Q2, CEX expenditure interviews have wave identifiers 2–5. From 2015Q2 onward, they have identifiers 1–4. We link observations only when the consumer-unit identifier is unchanged and both the interview wave and calendar quarter rise by one. This yields at most three one-quarter transitions per consumer unit.

Private income is the sum of reported labor, business, asset, pension, rental, and other private income. Public income is the sum of Social Security and other government transfers. We define post-tax income as private income plus public income minus reported taxes. Because the reported income amounts are annual, we divide them by four to express them at a quarterly rate. Total quarterly consumption expenditure is $TOTEXPPQ + TOTEXPCQ$.

We use two analysis samples. The private-income-ranked state and transition matrix use the income-complete sample. For this sample, the reported income components must sum to reported before-tax income. Private income, post-tax income, and consumption must be positive, and the survey weight must be valid and positive. We also exclude observations for which quarterly consumption exceeds 20 times quarterly post-tax income, equivalently five times reported annual post-tax income. The descriptive percentile series in Figure 2 use this income-complete sample under both rankings so that we can compare all three variables on a common set of records.

For the consumption-ranked DMD state and transition matrix, we use a broader sample that requires positive consumption and valid positive survey weights, including records with incomplete income data.

We convert nominal values to real values using the PCE price index. To remove the discontinuity associated with the CEX income redesign, we set the 2013Q2 level equal to the 2013Q1 level and then chain subsequent observed growth rates forward.

G Construction of the Dynamic Mode Decomposition

We record here the construction summarized in Section 4.5.

G.1 DMD reduction

Let n_y denote the dimension of the observable state, with $n_y = 1 + M$ under consumption ranking and $n_y = 1 + 3M$ under private-income ranking. If $y_1, \dots, y_\tau$ are the available state observations, we form

$$Z_- := [y_1, \dots, y_{\tau-1}], \quad Z_+ := [y_2, \dots, y_\tau].$$

We take the singular value decomposition $Z_- = U\Sigma V^\top$ and retain the leading N triplets (U_N, Σ_N, V_N) . The exact rank- N DMD estimator and its representation in the retained coordinates are

$$\hat{B} := Z_+ V_N \Sigma_N^{-1} U_N^\top, \quad \tilde{B} := U_N^\top \hat{B} U_N = U_N^\top Z_+ V_N \Sigma_N^{-1}. \quad (111)$$

In general, $\hat{B}$ differs from $U_N \tilde{B} U_N^\top$ because the range of the exact DMD operator is determined by Z_+ , whereas U_N comes from Z_- . We use $\hat{B}$ for the fitted full-state model and $\tilde{B}$ for the nonlinear value-function calculations.

Let

$$\hat{E} := Z_+ - \hat{B} Z_-, \quad \hat{\Omega} := \frac{\hat{E} \hat{E}^\top}{\tau - 1}, \quad \hat{\Gamma} = \hat{\Gamma} \hat{\Gamma}^\top, \quad (112)$$

where $\hat{\Gamma}$ is any covariance factor, including a rectangular factor when the covariance matrix is singular. The fitted full-state model is

$$y_{t+1} = \hat{B} y_t + \hat{\Gamma} \eta_{t+1}, \quad \eta_{t+1} \sim N(0, I_k). \quad (113)$$

Because $\hat{B}$ has rank at most N , this estimator imposes rank and subspace restrictions on the autoregressive law. Consistency for the population autoregressive matrix requires that the population dynamics satisfy these restrictions. We discuss rank selection in Appendix I.

G.2 Exact reduced Bellman equation for the fitted DMD model

Write $U := U_N$ and $K := Z_+ V_N \Sigma_N^{-1}$. The fitted rank- N estimator satisfies $\hat{B} = K U^\top$ identically. Thus, for $x_t := U^\top y_t$, the full fitted law and its reduced coordinates satisfy

$$y_{t+1} = K x_t + \hat{\Gamma} \eta_{t+1}, \quad (114)$$

$$x_{t+1} = \tilde{B} x_t + U^\top \hat{\Gamma} \eta_{t+1}, \quad \tilde{B} := U^\top K. \quad (115)$$

The conditional distribution of the entire next state depends on today's state only through x_t . The aggregate and percentile consumption equations retain their full conditional means:

$$\Delta Y_{t+1} = \nu + e_1^\top K x_t + e_1^\top \widehat{\Gamma} \eta_{t+1}, \quad (116)$$

$$\Delta c_{q,t+1} = \nu + (e_1^\top + e_q^{(c)}) K x_t - e_q^{(c)} y_t + (e_1^\top + e_q^{(c)}) \widehat{\Gamma} \eta_{t+1}. \quad (117)$$

The known current coordinate $e_q^{(c)} y_t$ enters consumption growth because growth subtracts current consumption. The terms $K x_t$ in (116)–(117) include output components outside the columns of U .

The current consumption level is $c_p(Y, y) = Y + \nu_p^{(c)} + e_p^{(c)} y$. Translation equivariance and uniqueness in the linear-growth class give $v_p(Y + b, y) = v_p(Y, y) + b$ for a common permanent shift b . If $U^\top h = 0$, states y and $y + h$ have identical next-state distributions under (114). Their continuation terms therefore agree, and the Bellman equation implies

$$v_p(Y, y + h) - v_p(Y, y) = (1 - \beta) e_p^{(c)} h, \quad U^\top h = 0.$$

After subtracting $Y + (1 - \beta) e_p^{(c)} y$, the remaining value is consequently constant on every set of states sharing the same $U^\top y$. Removing also the common trend constant gives the exact representation

$$v_p(Y, y) = Y + (1 - \beta) e_p^{(c)} y + \frac{\beta \nu}{1 - \beta} + \hat{v}_p(U^\top y). \quad (118)$$

For every choice of the two robustness penalties, $\hat{v}_p(x)$ contains all nonlinear dependence on the current state, and dependence on the discarded coordinates is affine.

Define $d_j^\top := e_1^\top + (1 - \beta) e_j^{(c)}$. Since $Y' = Y + \nu + e_1^\top y'$, the destination- j continuation value is

$$v_j(Y', y') = Y + \nu + \frac{\beta \nu}{1 - \beta} + d_j^\top y' + \hat{v}_j(x').$$

Substituting into (18), applying translation equivariance, and using $\beta[\nu + \beta\nu/(1 - \beta)] = \beta\nu/(1 - \beta)$ gives

$$\hat{v}_p(x) = (1 - \beta) \nu_p^{(c)} - \theta_\Pi \log \left[\sum_{j=0}^{M-1} \Pi_{pj} \exp \left(-\frac{(\mathcal{R}_\varepsilon \hat{v}_j)(x)}{\theta_\Pi} \right) \right], \quad (119)$$

where, before integrating out any innovation,

$$(\mathcal{R}_\varepsilon \hat{v}_j)(x) := -\theta_\varepsilon \log \mathbb{E}_\eta \exp \left[-\frac{\beta}{\theta_\varepsilon} \{ \hat{v}_j(x') + d_j^\top y' \} \right], \quad y' = Kx + \widehat{\Gamma} \eta, \quad x' = U^\top y'.$$

An infinite penalty is understood as its expectation limit. In particular, the outer continua-

tion term in (119) becomes $\sum_j \Pi_{pj}(\mathcal{R}_\varepsilon \hat{v}_j)(x)$ when $\theta_\Pi = \infty$.

To reduce the innovation integral, let $\Omega := \widehat{\Gamma} \widehat{\Gamma}^\top = \widehat{\Omega}$, $Q_x := U^\top \Omega U$, and $r := \text{rank}(Q_x)$. Choose a thin factor $L \in \mathbb{R}^{N \times r}$ of full column rank such that $LL^\top = Q_x$. The retained innovation is $w := U^\top \widehat{\Gamma} \eta = L\xi$, where $\xi \sim N(0, I_r)$. For the scalar innovation $v_j := d_j^\top \widehat{\Gamma} \eta$, define

$$q_j := U^\top \Omega d_j, \quad a_j := L^+ q_j, \quad \sigma_{\perp,j}^2 := d_j^\top \Omega d_j - \|a_j\|^2,$$

where L^+ is the Moore–Penrose pseudoinverse. Positive semidefiniteness of the joint covariance of (w, v_j) implies $q_j \in \text{range}(Q_x) = \text{range}(L)$, so $La_j = q_j$. Gaussian conditioning therefore yields

$$v_j = a_j^\top \xi + \sigma_{\perp,j} z, \quad z \sim N(0, 1), \quad z \perp \xi, \quad (120)$$

for each destination j , with conditional variance $\sigma_{\perp,j}^2 = \text{Var}(v_j \mid w) \geq 0$. We evaluate expectations conditional on the realized destination, allowing residuals across destinations to be correlated. The factorization applies to singular Q_x as well. When $r = 0$, the retained innovation and a_j are empty, and the remaining expectation is deterministic.

Integrating the independent scalar residual in (120) analytically gives

$$(\mathcal{R}_\varepsilon \hat{v}_j)(x) = \begin{cases} \beta d_j^\top Kx - \frac{\beta^2 \sigma_{\perp,j}^2}{2\theta_\varepsilon} \\ \quad - \theta_\varepsilon \log \mathbb{E}_\xi \exp \left[-\frac{\beta}{\theta_\varepsilon} \left\{ \hat{v}_j(\tilde{B}x + L\xi) + a_j^\top \xi \right\} \right], & \theta_\varepsilon < \infty, \\ \beta d_j^\top Kx + \beta \mathbb{E}_\xi \hat{v}_j(\tilde{B}x + L\xi), & \theta_\varepsilon = \infty. \end{cases} \quad (121)$$

The expectation has dimension $r \leq N$. The mean term $d_j^\top Kx$ retains all fitted output means, while $a_j^\top \xi$ preserves the correlation between the retained state and the payoff-relevant innovation. The scalar variance term integrates the independent conditional residual exactly.

For an affine reduced value $\hat{v}_j(x) = \gamma_j + \ell_j^\top x$, the complete continuation innovation has exposure $d_j + U\ell_j$ to the fitted residual. Since $La_j = q_j$, the reduced formula satisfies the identity

$$\begin{aligned} \|L^\top \ell_j + a_j\|^2 + \sigma_{\perp,j}^2 &= \ell_j^\top Q_x \ell_j + 2\ell_j^\top q_j + d_j^\top \Omega d_j \\ &= (d_j + U\ell_j)^\top \Omega (d_j + U\ell_j). \end{aligned} \quad (122)$$

Thus exact Gaussian evaluation reproduces the full-state Case 1 variance and intercept adjustment of Proposition 5. The affine Case 0 slopes, which are also the Case 1 slopes, can be computed directly in reduced coordinates. With row j of $\Lambda^{(N)}$ equal to $\ell_j^\top$, they satisfy

$$\Lambda^{(N)} = \beta \Pi \Lambda^{(N)} \tilde{B} + \beta \Pi D_K, \quad (D_K)_{j\cdot} := d_j^\top K. \quad (123)$$

The Case 0 intercept vector is $\gamma = (1 - \beta)(I - \beta\Pi)^{-1}\nu^{(c)}$, where $\nu^{(c)} := (\nu_0^{(c)}, \dots, \nu_{M-1}^{(c)})^\top$.

Finally, lifting a solution of (119)–(121) by (118) yields a fixed point of the full fitted Bellman equation, and every full fixed point in the linear-growth class has this representation. The lift is injective because $U^\top U = I_N$; evaluating at $y = Ux$ recovers $\hat{v}_p(x)$. It preserves the linear-growth class. Under the assumptions of Proposition 13, the unique full fixed point in that class therefore gives the unique reduced fixed point in the corresponding class.

H Numerical implementation details

H.1 Value-function iteration on the grid

For Cases 2 and 3, we solve (65) by value-function iteration on a regular grid in the N -dimensional reduced state space x . The grid spans ± 4 standard deviations of the stationary distribution of x in each coordinate, with 10 equally spaced points per dimension (10^5 grid points total for $N = 5$). We approximate expectations over $\xi \sim N(0, I_N)$ using the $2N+1$ sigma points when Q_x is nonsingular

$$\xi_0 = 0, \quad \xi_i^\pm = \pm \sqrt{N+1} e_i, \quad \omega_0 = \frac{1}{N+1}, \quad \omega_i^\pm = \frac{1}{2(N+1)}.$$

If $r = \text{rank}(Q_x) < N$, we replace N by r in this innovation rule and use the thin covariance factor L . For $r = 0$, the next reduced state is deterministic. These points reproduce the mean and covariance of the standard normal distribution. The numerical Bellman and detection experiments use this finite-support approximation for retained-state innovations. For a query x' outside the grid, we let $\bar{x}'$ be its coordinatewise projection onto the nearest grid boundary. We first apply multilinear interpolation at $\bar{x}'$ and then add the Case 0 affine extrapolation $\Lambda^{(N)}(x' - \bar{x}')$, where $\Lambda^{(N)}$ solves (123). Let $\hat{\Phi}$ denote the map on grid values defined by (65) after quadrature, off-grid affine extrapolation, and multilinear interpolation. We initialize $\hat{v}^{[0]} \equiv 0$ and iterate $\hat{v}^{[i+1]} = \hat{\Phi}\hat{v}^{[i]}$ until $\|\hat{v}^{[i+1]} - \hat{v}^{[i]}\|_\infty < 10^{-6}$.

The implemented operator is a contraction in the supremum norm, which justifies this stopping rule. The fixed extrapolation term cancels when two candidate grid functions are compared. Interpolation is monotone and reproduces constants, and both certainty-equivalent stages are increasing and translation equivariant. These are Blackwell's sufficient conditions for a contraction (Blackwell, 1965), also discussed by Stokey et al. (1989, Theorem 3.3) and Rust (1996). Hence, for grid values f and g with $r := \|g - f\|_\infty$, the pointwise

bounds $f - r \leq g \leq f + r$ propagate through $\widehat{\Phi}$ as $\widehat{\Phi}f - \beta r \leq \widehat{\Phi}g \leq \widehat{\Phi}f + \beta r$, so that

$$\|\widehat{\Phi}g - \widehat{\Phi}f\|_\infty \leq \beta \|g - f\|_\infty. \quad (124)$$

The implemented iteration therefore has a unique fixed point $\hat{v}^*$ on the grid, and the successive-iterate difference bounds the remaining error,

$$\|\hat{v}^{[i+1]} - \hat{v}^*\|_\infty \leq \frac{\beta}{1 - \beta} \|\hat{v}^{[i+1]} - \hat{v}^{[i]}\|_\infty. \quad (125)$$

At $\beta = 0.99$, the tolerance 10^{-6} puts the accepted iterate within 9.9×10^{-5} of the fixed point of $\widehat{\Phi}$ in the supremum norm.

H.2 Likelihood ratios used in calibration

We write

$$\hat{v}_{jn}(x) := \hat{v}_j(\widetilde{B}x + L\xi_n).$$

In the calibration experiment, we approximate the Gaussian innovation to the retained state using the sigma-point rule with probabilities $\{\omega_n\}$ from Appendix H.1. For the numerical Cases 2 and 3, the Bellman equation and the likelihood-ratio simulation use the same mixed finite-support–Gaussian innovation law, with sigma points for the retained state and the conditionally independent Gaussian residual from Appendix G.2. For Case 1, we calibrate θ_ε with the same finite-support experiment and compute the reported welfare value from the continuous-Gaussian full-state formula in Proposition 5.

When $\theta_\varepsilon < \infty$, the distorted probability of retained-state innovation node n , conditional on next-period percentile j , is

$$\widetilde{\omega}_n^{(j)}(x) = \frac{\omega_n \exp\{-\beta[\hat{v}_{jn}(x) + a_j^\top \xi_n]/\theta_\varepsilon\}}{\sum_m \omega_m \exp\{-\beta[\hat{v}_{jm}(x) + a_j^\top \xi_m]/\theta_\varepsilon\}}. \quad (126)$$

The term $a_j^\top \xi_n$ preserves the dependence between the retained innovation and the innovation to the explicit linear part of the continuation payoff. The independent conditional residual is $\sigma_{\perp,j}z$, where $z \sim N(0, 1)$ under the approximating law. Define $\mathbf{m}_{\perp,j} := \beta\sigma_{\perp,j}/\theta_\varepsilon$. The standardized residual has worst-case law $N(-\mathbf{m}_{\perp,j}, 1)$ and likelihood ratio

$$g_{\perp,j}(z) = \frac{dN(-\mathbf{m}_{\perp,j}, 1)}{dN(0, 1)}(z) = \exp\{-\mathbf{m}_{\perp,j}z - \tfrac{1}{2}\mathbf{m}_{\perp,j}^2\}. \quad (127)$$

The transition tilt uses the grid certainty equivalent $(\widehat{\mathcal{R}}_\varepsilon \hat{v}_j)(x)$ in (66), including both the

correlated innovation term and the analytically integrated conditional residual variance. When $\theta_\varepsilon = \infty$, (126) is replaced by $\tilde{\omega}_n^{(j)} = \omega_n$ and $g_{\perp,j} = 1$, and (66) takes its expectation limit. If $\sigma_{\perp,j} = 0$, the residual statistic is omitted and its likelihood contribution is one.

For a current percentile p , the distorted transition probabilities are

$$\tilde{\Pi}_{pj}(x) = \frac{\Pi_{pj} \exp\{-(\hat{\mathcal{R}}_\varepsilon \hat{v}_j)(x)/\theta_\Pi\}}{\sum_k \Pi_{pk} \exp\{-(\hat{\mathcal{R}}_\varepsilon \hat{v}_k)(x)/\theta_\Pi\}}. \quad (128)$$

When $\theta_\Pi = \infty$, $\tilde{\Pi}_{pj}(x) = \Pi_{pj}$. For a simulated one-period outcome $(p, x) \rightarrow (j, n, z)$, the one-step log likelihood ratio is

$$\Delta\mathcal{L}(p, x; j, n, z) = \log \frac{\tilde{\Pi}_{pj}(x)}{\Pi_{pj}} + \log \frac{\tilde{\omega}_n^{(j)}(x)}{\omega_n} + \log g_{\perp,j}(z). \quad (129)$$

Case 1 sets the first term to zero, and Case 2 sets the second and third terms to zero. Case 3 retains all three. Estimated zero transition cells remain outside the support, so a ratio in the first term is evaluated only when $\Pi_{pj} > 0$.

H.3 Initial state, sample likelihood, and search

For paths generated under either model, we use the same initial distribution. We draw p_0 from the stationary distribution of the approximating transition matrix and, independently, draw $x_0 \sim N(0, \Sigma_x)$, where

$$\Sigma_x = \tilde{B}\Sigma_x\tilde{B}^\top + LL^\top.$$

The common initial distribution cancels from the likelihood ratio, and we begin recording transitions immediately. The simulated sample log-likelihood ratio is

$$\mathcal{L}_T = \sum_{t=0}^{T-1} \Delta\mathcal{L}(p_t, x_t; p_{t+1}, n_{t+1}, z_{t+1}), \quad T = 137. \quad (130)$$

We set the calibration horizon to $T = 137$, the number of quarterly dates from 1990Q1 through 2024Q1. The fitted DMD law uses 135 adjacent state transitions because constructing the state loses the first date.

For Case 1, the full Gaussian model also supplies an exact calibration check. Conditional on the percentile path, put $S_T = (\beta^2/\theta_\varepsilon^2) \sum_{t=1}^T \mathbf{e}_{p_t}$, where $\mathbf{e}_j = \|s_j\|^2$ is the full-state continuation-value variance. Then $\mathcal{L}_T$ is $N(-S_T/2, S_T)$ under Q and $N(S_T/2, S_T)$ under $\tilde{Q}$.

The percentile path has the same law under both models, so

$$\bar{p}_1(\theta_\varepsilon) = \mathbb{E}_{p_{0:T}} \left[\Phi_N \left(-\frac{\beta}{2\theta_\varepsilon} \sqrt{\sum_{t=1}^T \mathbf{e}_{p_t}} \right) \right], \quad p_0 \sim \pi, \quad p_{t+1} \mid p_t \sim \Pi_{p_t}, \quad (131)$$

where Φ_N is the standard normal distribution function. The conditional detection error is $1/2$ when $S_T = 0$. This identity provides a continuous-Gaussian check on the finite-support calibration experiment.

We can derive a $T^{-1/2}$ rate in the equal-variance Gaussian benchmark of Remark 9. In that case, the Case 1 worst-case innovation has mean shift $\mathbf{m}_j = -(\beta/\theta_\varepsilon)s_j$, with a common norm across percentiles. The T innovation likelihood contributions imply

$$\bar{p} = \text{Prob} \left(Z > \frac{1}{2} \sqrt{T} \|\mathbf{m}\| \right), \quad Z \sim N(0, 1).$$

Holding $\bar{p}$ fixed therefore makes $\|\mathbf{m}\|$, and hence the Case 1 welfare increment $-\kappa$, decline at rate $T^{-1/2}$.

Algorithm 1: Detection-error calibration for one robust case

Input: Case, target $\bar{p}_{\text{target}}$, bracket for θ , common seeds

- 1 **repeat**
- 2 Set $\theta = \sqrt{\theta_L \theta_U}$ for the current positive bracket $[\theta_L, \theta_U]$
- 3 Solve the corresponding Bellman equation
- 4 Construct (126), (127), and (128)
- 5 Simulate paths under Q and $\tilde{Q}$ from the common initial convention
- 6 Accumulate all 137 terms in (130)
- 7 Compute $p_A = \Pr_Q(\mathcal{L}_T > 0)$, $p_B = \Pr_{\tilde{Q}}(\mathcal{L}_T \leq 0)$, and $\bar{p} = (p_A + p_B)/2$
- 8 Increase θ if $\bar{p} < \bar{p}_{\text{target}}$ and decrease it otherwise
- 9 **until** the log-penalty bracket is sufficiently narrow

Output: Calibrated penalty parameter, achieved $\bar{p}$, and simulation record

We simulate 2,000 paths under each model and use common random numbers across candidate penalties. During calibration, we stop value-function iteration when the successive-iterate difference is below 10^{-5} . For the final reported values, we use 10^{-6} . Near $\bar{p} = 0.45$, 2,000 paths under each model give the averaged detection-error estimate an approximate Monte Carlo standard error of 0.0079, conditional on the fitted model.

To average welfare over the stationary reduced state, we evaluate the stationary Gaussian density at each node of the regular grid and normalize the resulting weights to sum to one. The resulting density-weighted sum approximates the Gaussian integral using only grid nodes. If Σ_x is singular, we integrate on its Gaussian support. We use affine extrapolation

outside the grid only when the Bellman equation or likelihood-ratio simulation queries an off-grid state.

I DMD rank sensitivity

Figure 7 reports two diagnostics for each DMD truncation rank N . The first is the share of the squared Frobenius norm of Z_- accounted for by the first N singular values $\varsigma_1 \geq \varsigma_2 \geq \dots$,

$$\frac{\sum_{j=1}^N \varsigma_j^2}{\sum_j \varsigma_j^2}.$$

The second is the expanding-window, one-step-ahead forecast RMSE for the consumption coordinates of the observable state y_{t+1} . These coordinates are the percentile-specific consumption deviations in (48) and (49). The forecast error for consumption growth $\Delta c_{q,t+1}$ also includes the error in the first coordinate of y_{t+1} .

Let $T_{\text{DMD}} := \tau - 1$ denote the number of adjacent observation pairs in (Z_-, Z_+) . We set $s_{\min} = \lfloor T_{\text{DMD}}/2 \rfloor$. At forecast origin s , we estimate the rank- N DMD operator from the first s pairs,

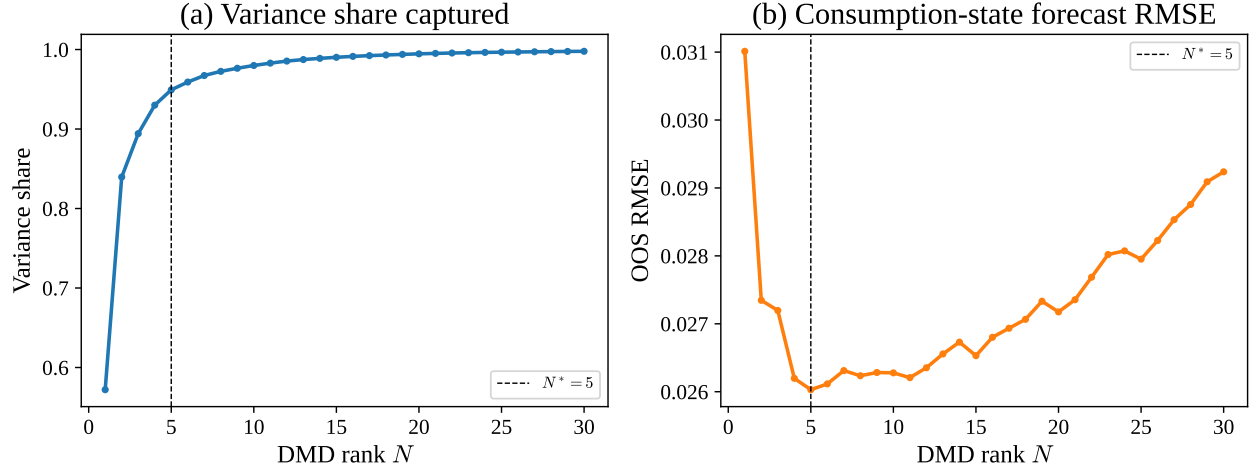
$$(y_1, y_2), \dots, (y_s, y_{s+1}),$$

and form the one-step-ahead forecast

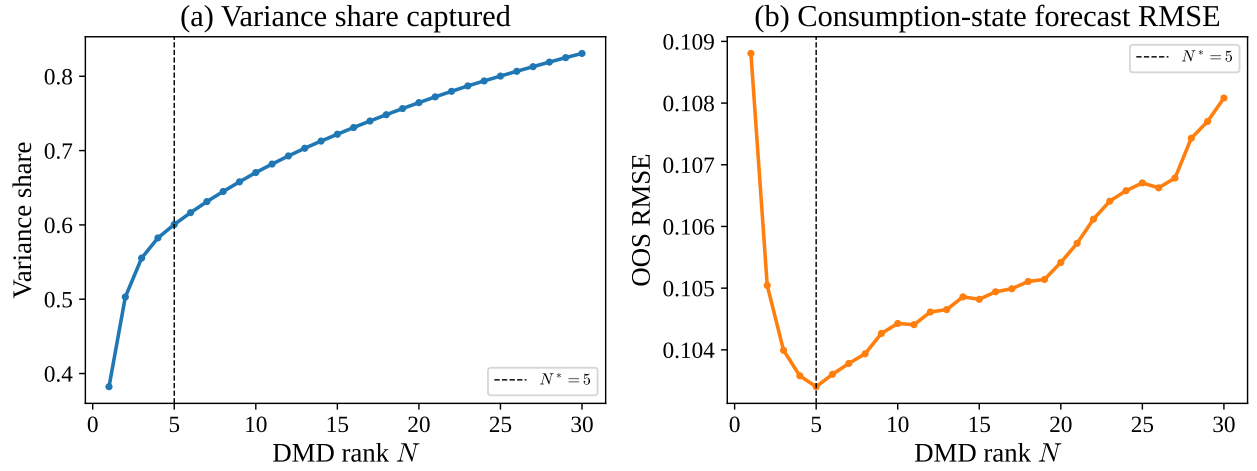
$$\hat{y}_{s+2} = \hat{B}^{(s)} y_{s+1}.$$

We repeat this calculation for $s = s_{\min}, \dots, T_{\text{DMD}} - 1$ and compute the RMSE over the consumption coordinates and all $T_{\text{DMD}} - s_{\min}$ forecast dates.

Under consumption ranking, this RMSE is minimized at $N = 5$, with values of 0.026030 at $N = 5$ and 0.026115 at $N = 6$. Under private-income ranking, it is also minimized at $N = 5$, with values of 0.103402 at $N = 5$ and 0.103603 at $N = 6$. The differences between the fifth and sixth ranks are small. We use $N = 5$ in both specifications.



(a) Consumption ranking.



(b) Private-income ranking.

Figure 7: Sensitivity to the DMD truncation rank. Within each ranking, the left panel reports the share of the squared Frobenius norm of Z_- accounted for by the retained singular values. The right panel reports the expanding-window, one-step-ahead RMSE for the consumption coordinates of y_{t+1} . Dashed lines mark the rank that minimizes this RMSE.

J The joint increment at a common detection-error target

At the target $\bar{p} = 0.45$, we compare the Case 3 increment over Case 0 with the sum of the separately calibrated Case 1 and Case 2 increments using the ratio

$$\frac{\bar{\alpha}_p^{(3)} - \alpha_p^{(0)}}{(\alpha_p^{(1)} - \alpha_p^{(0)}) + (\bar{\alpha}_p^{(2)} - \alpha_p^{(0)})}. \quad (132)$$

A ratio of one means the joint increment equals the sum of the separate increments, and a ratio below one means the joint increment falls short of that sum. The conversion factor of Section 4.7 is positive and common to every compensating difference, so the ratio is the same whether it is formed from $\bar{\alpha}_p^{(m)}$ or from the annualized entries of Table 3. At the 10th consumption percentile, the joint increment is 0.1350 annual percentage points and the separate increments sum to 0.2037, giving a ratio of 0.663. Across the three reported percentiles, the ratio is about 0.663 under consumption ranking and 0.588–0.589 under private-income ranking.

The cases reach the same detection-error target at different penalties. In Cases 1 and 2, a single distortion attains the detection-error target. In Case 3, both distortions enter the likelihood ratio (62). The calibrated common penalty therefore restrains each distortion more than its separate calibration does.

To measure this effect, we solve an auxiliary joint problem, Case 4, using the Case 1 innovation penalty and the Case 2 transition penalty. These are $\theta_\varepsilon = 1.0288$ and $\theta_\Pi = 0.9268$ under consumption ranking, and $\theta_\varepsilon = 0.8466$ and $\theta_\Pi = 0.7953$ under private-income ranking. The resulting joint detection-error probabilities are 0.4253 and 0.4190, respectively, below the target of 0.45.

At these separate penalty parameters, the joint increment is 99.6–99.7 percent of the sum of the separate increments at the reported percentiles under both rankings. The 10th-percentile cost rises to 0.2886 under consumption ranking and 0.2893 under private-income ranking. Both exceed the corresponding Case 3 costs, 0.2207 and 0.2139. This is expected since the two components in Case 3 are constraint by the same entropy penalty parameter, which restricts the distortion compared to the case here where we allow for separate penalty terms.

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